

Holonomic Rigidity for Entire Hahn Functions

A sharp order-unit criterion for dilations,
with uniform exclusion bounds,
and first-order nonlinear rigidity

Merged research report · 22 September 2026

Abstract

Fix a nonzero set-sized ordered abelian group Γ and the complex Hahn field $K_\Gamma = \mathbb{C}((t^\Gamma))$. *Entire* means strong Hahn summability of the evaluation family at every point of this one field. We prove that every strongly entire D-finite power series is a polynomial, in one or in finitely many variables, and that a fixed differential operator already excludes every nonterminating formal solution on an explicit exterior valuation region, with a common degree bound.

For dilations there is an exact alternative. If $q \in K_\Gamma^\times$ has infinite multiplicative order, a nonpolynomial strongly entire solution of a nonzero linear q -difference equation with polynomial coefficients exists if and only if $v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ . A nonzero valuation is not enough: its integer multiples must be cofinal among all scales. The proof covers units whose residues are roots of unity, and coefficients and parameters with arbitrary well-ordered Hahn supports. A partial theta series proves sharpness, and its exact strong evaluation domain is computed for an arbitrary Hahn parameter. Quantitatively, the recurrence supplies an exclusion bound that is attained on its own boundary, and a coarsened exclusion bound valid for every noncofinal dilation. A mixed differential–dilation extension is proved for nonzero noncofinal dilation valuations.

The common mechanism is an elementary finite-step escape chain: bounded valuation costs in a coefficient recurrence force a subsequence of evaluated leading exponents that never increases. This contradicts strong summability without replacing an arbitrary-rank valuation by a real-valued norm. In an explicit surreal workspace with countable cofinality and no order unit, the series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is strongly entire yet satisfies no nonzero linear differential equation and no nonzero linear dilation equation for any nontorsion parameter of that workspace.

A third source adds a nonlinear part over $k((t^\Gamma))$, for any field k of characteristic zero: every strongly entire solution of a nonzero first-order algebraic differential equation $P(z, f, D_z f) = 0$ is a polynomial, with an exclusion bound that must depend on the solution. Higher order stays open outside a nondegenerate class.

Γ is *not* assumed divisible. Divisibility is hypothesised only where it is used, in the three threshold constructions listed in Section 1.4, and each of those statements prints the hypothesis itself.

Status. An AI-assisted research report; priority and independent refereeing are not certified. Lean covers the positive-support word lemma, escape mechanism, bounded-cost exclusion and formal exponential domain. The main differential and dilation classifications, the new torsion argument and the whole nonlinear part remain unformalized. The variable derivative is not the Berarducci–Mantova derivation on surreal coefficients.

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1 The question and the principal results

1.1 Why the question is scale sensitive

A familiar way to specify a special function is to give a finite linear differential equation. Another is to give a finite relation between its values at $z, qz, q^2z, \dots$. Over the complex numbers both descriptions produce nonpolynomial entire functions. Over a surreal Hahn workspace, “entire” asks for evaluation not only at ordinary complex constants but at every infinite and infinitesimal scale available in that workspace. In a Hahn field an infinite expression is legal only when its supports satisfy an order condition and a finiteness condition, and an equation can force either condition to fail at a sufficiently large argument. The question of this report is what a finite linear equation can accommodate under that stronger global requirement. Sections 14–17 ask the same of a finite *nonlinear* algebraic differential equation, and answer it in order one.

The answer depends decisively on which operator is used, and for dilation equations it depends further on whether the valuation of the particular parameter reaches every scale of the value group.

1.2 Conventions

We work with a fixed ordered abelian group Γ and

$$K_\Gamma = \mathbb{C}((t^\Gamma)).$$

An element of K_Γ is a formal sum $a = \sum_\gamma a_\gamma t^\gamma$ with $a_\gamma \in \mathbb{C}$ whose support $\text{supp}(a) = \{\gamma : a_\gamma \neq 0\}$ is a well-ordered set. For $a \neq 0$ put

$$v(a) = \min \text{supp}(a), \quad \text{lc}(a) = a_{v(a)} \in \mathbb{C}^\times, \quad \text{res}(a) = \text{lc}(a) \text{ when } v(a) = 0,$$

and put $v(0) = +\infty$ only where convenient; this symbol is never subtracted. Products and sums are the usual Hahn operations, with

$$v(ab) = v(a) + v(b), \quad v(a + b) \geq \min\{v(a), v(b)\}, \tag{1.1}$$

with equality in the second law when the two valuations differ. The coefficient field $\mathbb{C}$ is trivially valued, so every nonzero ordinary integer and every factorial has valuation zero. Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$, and $|\lambda|$ means $\max(\lambda, -\lambda)$ in the ordered group.

Convention 1.1 (Standing hypothesis on the value group). Γ is a nonzero set-sized ordered abelian group. Divisibility is *not* part of this standing hypothesis. Every statement below that needs divisibility says so in its own statement; those statements are collected in Section 1.4.

Definition 1.2 (Strong summability, evaluation domain, entireness). A set-indexed family $(b_i)_{i \in I}$ in K_Γ is *strongly Hahn summable* if $\bigcup_i \text{supp}(b_i)$ is well ordered and each $\gamma \in \Gamma$ lies in the supports of only finitely many members. Its sum is then defined coefficientwise, by a finite sum in $\mathbb{C}$ at each exponent. For $f = \sum_{n \geq 0} a_n z^n \in K_\Gamma[[z]]$ let $\text{Dom}_{\text{str}}(f)$ be the set of $x \in K_\Gamma$ for which $(a_n x^n)_{n \geq 0}$ is strongly summable, and let

$$\mathcal{E}_\Gamma = \{f : \text{Dom}_{\text{str}}(f) = K_\Gamma\}.$$

In d variables the family to be tested is $(a_\alpha x^\alpha)_{\alpha \in \mathbb{N}^d}$.

The word *entire* always means Definition 1.2 relative to one fixed workspace. It does not mean that a complex coefficient function is entire, it does not mean convergence in the fine topology inherited from the full surreal class, and it is never a claim about all of $\mathbf{No}[i]$. The definition also concerns the family *before* cancellation between different indices: one cannot first cancel a nonsummable family and then declare it summable, and convergence of a numerical coefficient series does not license infinitely many contributions at one Hahn exponent.

Definition 1.3 (Order unit). A positive $\mu \in \Gamma$ is an *order unit* if for every $\gamma \in \Gamma$ there is $N \in \mathbb{N}$ with $|\gamma| \leq N\mu$; equivalently, if the positive integer multiples of μ are cofinal in Γ . Zero is not an order unit.

Definition 1.4 (The two operator classes). Let D_z be the formal derivative in z , so $D_z(a) = 0$ for every $a \in K_\Gamma$, and for $q \in K_\Gamma^\times$ let $\sigma_q f(z) = f(qz)$. A series $f \in K_\Gamma[[z]]$ is *D-finite over $K_\Gamma(z)$* if

$$\sum_{r=0}^R p_r(z) D_z^r f(z) = 0, \quad p_r \in K_\Gamma[z], \quad (p_0, \dots, p_R) \neq 0, \quad (1.2)$$

and *linearly q -difference finite* if

$$\sum_{\ell=0}^r p_\ell(z) f(q^\ell z) = 0, \quad p_\ell \in K_\Gamma[z], \quad (p_0, \dots, p_r) \neq 0. \quad (1.3)$$

These are formal identities in $K_\Gamma[[z]]$, not assertions about limits of partial sums. Rational coefficients give the same notions after denominators are cleared.

We say “linearly q -difference finite” rather than silently identifying several meanings of “ q -holonomic”. In the sequence literature q is often an indeterminate with coefficients in $\mathbb{Q}(q)$; here q is a *fixed* Hahn element, possibly with infinitely many terms. The differential terminology is the usual characteristic-zero one [1]. *Nontorsion* and “of infinite multiplicative order” are used interchangeably.

1.3 The principal results

Theorem A (Differential rigidity, with a uniform exterior obstruction). *Let Γ be as in Convention 1.1 and let $0 \neq L = \sum_{r,k} c_{r,k} z^k D_z^r$ be a nonzero linear differential operator with coefficients in $K_\Gamma[z]$. There are $B_L \in \Gamma$ with $B_L \geq 0$ and $N_L \in \mathbb{N}$, both determined by the finitely many coefficients of L , such that every formal solution of $Lf = 0$ that is strongly evaluable at even one $x \neq 0$ with $v(x) \leq -B_L$ has degree less than N_L . Consequently*

$$\{f \in \mathcal{E}_\Gamma : f \text{ is } D\text{-finite over } K_\Gamma(z)\} = K_\Gamma[z].$$

Theorem B (Sharp dilation criterion). *Let Γ be as in Convention 1.1 and let $q \in K_\Gamma^\times$ have infinite multiplicative order. The following are equivalent.*

- (i) *There is a nonpolynomial $f \in \mathcal{E}_\Gamma$ that is linearly q -difference finite over $K_\Gamma(z)$.*
- (ii) *$v(q) \neq 0$ and $|v(q)|$ is an order unit of Γ .*

When (ii) fails, every fixed equation (1.3) has a uniform exterior obstruction and a uniform polynomial degree bound, exactly as in Theorem A. When (ii) holds, an explicit partial theta series with parameter q or q^{-1} is a nonpolynomial entire solution.

Theorem C (Several-variable rigidity). *Let Γ be as in Convention 1.1 and let $F \in K_\Gamma[[z_1, \dots, z_d]]$ be strongly Hahn summable at every point of K_Γ^d . If the mixed partial derivatives of F span a finite-dimensional vector space over $K_\Gamma(z_1, \dots, z_d)$, then F is a polynomial.*

The fourth principal theorem comes from the third source manuscript and is proved in Sections 14–15. It allows an arbitrary coefficient field of characteristic zero; see Convention 14.1, which also explains why that generality is not transferred to Theorems A–C.

Theorem D (First-order nonlinear rigidity). *Let Γ be as in Convention 1.1, let k be any field of characteristic zero, and let $\mathbb{K} = k((t^\Gamma))$, with strong evaluation and entireness defined as in Definition 1.2 with k in place of $\mathbb{C}$. For every nonzero $P \in \mathbb{K}[z, Y_0, Y_1]$, every strongly entire formal solution $f \in \mathbb{K}[[z]]$ of*

$$P(z, f, D_z f) = 0$$

is a polynomial. More strongly, if a nonpolynomial formal solution exists, there is $\delta_0 \in \Gamma_{>0}$, depending on the equation and on finitely many coefficients of that solution, such that $t^{-\delta} \notin \text{Dom}_{\text{str}}(f)$ for every $\delta \geq \delta_0$. Consequently, for every nonpolynomial strongly entire f , the series f and $D_z f$ are algebraically independent over $\mathbb{K}(z)$.

Theorem D has no separant, irreducibility or solved-form hypothesis on P . Its exclusion bound depends on the solution, and Theorem 15.10 shows that this cannot be avoided, in contrast with the operator-only bound of Theorem A. It does not say that a nonpolynomial entire function is differentially transcendental in all orders: independence of f and $D_z f$ does not exclude a relation involving $D_z^2 f$. Question 19.1 is re-scoped accordingly.

The hypotheses separate two obstructions. Differential equations have bounded valuation costs after passing to their coefficient recurrence. Dilation equations can have costs increasing linearly with the index, but only along the single scale $v(q)$; a scale that is not an order unit cannot reach all of Γ , even if its multiples are unbounded within their own Archimedean component. Strong entireness detects that failure.

Beyond the first three, the report proves: a mixed differential–dilation rigidity theorem at every nonzero noncofinal dilation scale (Theorem 10.2); a recurrence exclusion bound that is *attained*, boundary included, by an explicit formal exponential (Corollary 3.5 and Proposition 5.3); a coarsened exclusion bound valid at every nonzero noncofinal dilation valuation (Theorem 7.4); and the exact strong evaluation domain of the partial theta series for an arbitrary Hahn parameter, including arbitrary tails (Theorem 8.1).

1.4 Where divisibility of Γ is used

The core arguments — the escape mechanism, the orbit valuation lemmas, the two conversions, the theta domain theorem, and all of Theorems A–C — use only Convention 1.1. Divisibility of Γ enters in exactly three threshold constructions, with the hypothesis stated where it is used.

- (1) The refined recurrence threshold B_{rec} of (3.4), whose defining maximum divides a valuation difference by the jump length j . Corollary 3.5 states it under divisibility; Corollary 3.4 is the divisibility-free predecessor, with a coarser threshold and the same qualitative conclusion.
- (2) The refined coarsened threshold $\overline{B}_{\text{rec}}$ of (7.4), for the same reason, in the quotient group. Theorem 7.4 states the divisibility-free form and the divisible refinement separately.
- (3) The worked sparse threshold in Example 13.1, an instance of (1).

The torsion eigenspace equivalence in Proposition 9.1 needs no divisibility: inward stability of strong evaluation replaces the root-extraction argument of the source manuscripts.

The nonlinear part, Sections 14–17, uses no divisibility anywhere: its only divisions are of ordinary integers in $\mathbb{Q}$ (the degree cutoff (15.11)), and its exclusion bounds are chosen from finitely many strict inequalities with positive integer multipliers.

No theorem in this report states a conclusion under Convention 1.1 while resting on a proof step that divides by an integer in Γ . The division-free predecessors in (1) and (2) are weaker in the constant only, not in the conclusion.

1.5 Position relative to the collection and the literature

Provenance. This report is built from three manuscripts, numbered as they arrived in the collection. Sources 08 and 09 were written independently on 22 September 2026 and both prove the differential rigidity theorem and the order-unit dilation criterion (Theorems A and B); source 08 is the base, because it assumes only $\Gamma \neq 0$ while source 09 assumed divisibility, and every proof taken from 09 was re-read for a hidden division (Section 1.4). Together they supply Sections 1–13. Source 10, delivered later the same day, supplies the nonlinear part, Sections 14–17 and Theorem D. The repository snapshot consulted by sources 08 and 09 is

a3124af79f66b8b9c196d76b4cbc5ac3938907c4,

and the one consulted by source 10 is

048b72cf7cbfc8ab246e4f73788c10460cb3f6e0.

At the second pin this report consisted of sources 08 and 09 alone and its text was identical to the text into which source 10 was merged, so source 10’s references to “Question 15.1” and “Theorems A–C” of this report were accurate there; that question is now Question 19.1, re-scoped in Section 19. The choices the merge made (renamed symbols, results printed once, the placement of the nonlinear part after the linear theory) are recorded in Section 14.1.

The companion entire-function report. The collection already contains a merged report on cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank [12]. It fixes the same class $\mathcal{E}_\Gamma$ over the same kind of workspace, classifies the ring structure of $\mathcal{E}_\Gamma$, and computes the exact strong evaluation domain after an enlargement of the value group. None of its ring-theoretic conclusions is used to prove anything here: the present proofs start from Hahn support arithmetic. Two points of contact matter.

First, we do not reprove the order-unit coarsening machinery. That report’s Lemma 9.1, labelled `ent:lem:coarsening`, states that if $h > 0$ is an order unit of Γ there is an order-preserving homomorphism $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ whose kernel is the convex subgroup $H = \{\gamma : n|\gamma| < h \text{ for all } n \geq 1\}$, so that $|a|_\pi = \exp(-\pi(v(a)))$ is a nontrivial real-valued non-Archimedean absolute value inducing the intrinsic valuation topology. That is exactly the sense in which sources 08 and 09 of this report use the phrase “coarsening”, and we cite it rather than repeat it; see Remark 2.6.

Second, and this is easy to get wrong: *the two order-unit dichotomies are not the same theorem* — the companion report’s order unit decides Hermite interpolation and the Bézout property of the ring $\mathcal{E}_\Gamma$ (its Theorem 9.3), whereas the order unit here decides which annihilating operators a nonpolynomial member of $\mathcal{E}_\Gamma$ can satisfy. Neither dichotomy implies the other, and the coincidence

of the invariant is not a coincidence of statement. In particular the existence of nonpolynomial entire functions at countable cofinality, and the fact that $\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q}\omega^j$ has countable cofinality and no order unit, are background from that report [12], not targets of this one.

The collection’s differential-equations report uses a derivation on surreal *scalars* [13]; here D_z kills every scalar and differentiates only the formal variable, and no theorem about that scalar derivation is restated as ordinary D-finite rigidity. The rank-one analytic report [14] already contains the quadratic-exponent example $\sum t^{n^2} z^n$; our partial theta witness is explicitly classical and is not counted as a new function.

External literature. Stanley’s D-finite/P-recursive correspondence is the standard algebraic starting point [1]. We make the coefficient conversion explicit because the valuations of the resulting recurrence, rather than ordinary coefficient growth, drive the proof. Ramis treats growth of ordinary complex entire solutions of linear q -difference equations and also studies simultaneous differential and difference constraints [2]; the arbitrary-rank strong summability condition here is a different global condition, and none of its order-unit conclusions is inferred from an ordinary complex growth estimate. Garoufalidis proves eventual quadratic quasi-polynomial degree behaviour for q -holonomic sequences in a rational-function and Puiseux setting [3]; that is an important antecedent to the idea that dilation equations constrain coefficient scales, we do not claim to originate it, and we do not strengthen its quadratic quasi-polynomial conclusion. Di Vizio’s ultrametric Maillet–Malgrange theorem treats Gevrey behaviour of formal solutions of nonlinear ultrametric q -difference equations, including the valuation-zero case with Diophantine conditions [4]; we make no nonlinear generalization of it. The partial theta series belongs to the classical partial-theta literature [5], and Higman’s finite-word theorem is a standard external ingredient in the positive-support monoid lemma [6].

Terminology needs care in two further directions. Hahn meromorphic functions in the analytic setting of Müller and Strohmaier involve a convergence framework different from the formal strong evaluation used here [7]. The Berarducci–Mantova derivation differentiates surreal coefficients themselves and is not D_z [8]. Work on operation-closed surreal subfields gives context for set-sized workspaces [9], and the Conway normal-form conventions used in Section 2.4 follow current surreal field work [10]; closure under a scalar derivation is assumed in no theorem below.

The inspections of sources 08 and 09 were targeted: repository documentation, the report catalogue, the entire-function report, and the primary sources named above. Neither was a line-by-line audit of every repository manuscript or an exhaustive bibliography search, and a repository content search for `holonomic` returning no matches is not proof of absence. Source 10’s repository and literature comparison, equally targeted, is recorded in Section 17.4. Section 19 records exactly what is and is not claimed.

2 Hahn support and what evaluation means

2.1 Support calculus

Lemma 2.1 (Hahn–Neumann support calculus and the positive monoid). *Let A, B be well-ordered subsets of an ordered abelian group.*

- (1) $A + B$ is well ordered, and each γ has only finitely many representations $\gamma = a + b$ with $(a, b) \in A \times B$.

(2) If S is a well-ordered subset of $\Gamma_{>0}$, the additive monoid

$$M(S) = \{s_1 + \cdots + s_k : k \geq 0, s_i \in S\}$$

is well ordered, and each of its elements has only finitely many representations as a finite word in letters from S .

(3) Multiplication by a fixed Hahn element preserves strong summability, and multiplication by a nonzero one preserves and reflects it.

Proof. An infinite sequence in a well-ordered set has an infinite nondecreasing subsequence. Given a strictly decreasing sequence $a_n + b_n$, pass first to a subsequence with a_n nondecreasing and then to one with b_n nondecreasing, a contradiction. If infinitely many distinct pairs sum to a fixed γ , the distinct first coordinates contain an increasing sequence, forcing a decreasing sequence of second coordinates. This is part (1).

For part (2), order finite words over S by subsequence embedding with coordinatewise increase. Higman's theorem says that words over a well-quasi-ordered alphabet are well quasi ordered [6]. An embedding cannot decrease the sum, and increases it strictly unless the two words are identical, since inserted letters are positive and a strict coordinate increase is positive. A descending sequence of sums therefore gives a bad sequence of words, and so do infinitely many distinct words with one fixed sum. Both are impossible.

For part (3), let A be the union of the supports of a strongly summable family and B the support of the multiplier. The new supports lie in $A + B$. Part (1) leaves finitely many admissible pairs at a specified exponent, and for each first coordinate only finitely many family members contribute. For a nonzero multiplier apply the same to its inverse. $\square$

In part (2) words carry their order; a fixed finite multiset has only finitely many permutations, so the commutative monoid version follows. The argument does not assume that S is bounded below by an order unit, and remains valid at arbitrary valuation rank.

Lemma 2.2 (The nonincreasing leading-exponent obstruction). *A family of nonzero Hahn elements possessing an infinite subsequence whose leading exponents are nonincreasing is not strongly summable.*

Proof. If that sequence of leading exponents takes infinitely many distinct values it has an infinite strictly decreasing subsequence, so the union of the supports is not well ordered. Otherwise it is eventually constant, and one exponent lies in the supports of infinitely many members, so local finiteness fails. $\square$

The second alternative is essential and is what makes the exclusion boundary below a *closed* region. Merely excluding a descending support does not establish strong summability: the family $(1 \cdot z^n)_{n \geq 0}$ at $z = 1$ has support union $\{0\}$, yet every summand contributes at exponent zero.

Lemma 2.3 (Two sufficient certificates). *Let $(b_n)_{n \geq 0}$ be a family in K_Γ .*

- (a) *If $v(b_n)$ tends cofinally to $+\infty$, meaning that for each $\beta \in \Gamma$ all but finitely many nonzero b_n satisfy $v(b_n) > \beta$, then (b_n) is strongly summable.*
- (b) *If $\text{supp}(b_n) \subseteq h_n + M(S)$ for a well-ordered $S \subseteq \Gamma_{>0}$, where the set of values of h_n is well ordered and each value is taken by only finitely many indices, then (b_n) is strongly summable.*

Moreover, grouping a strongly summable family into finite blocks preserves strong summability and its sum.

Proof. For (a) fix β . Only finitely many supports meet $(-\infty, \beta]$, and a finite union of well-ordered subsets of a linear order is well ordered. A descending sequence in the full union would lie below its first member, hence inside such a finite union. The same finiteness bounds the multiplicity of an exponent. For (b) use parts (1) and (2) of Lemma 2.1: a fixed target exponent admits finitely many pairs $(h, m) \in \{h_n\} \times M(S)$ with $h + m$ equal to it, and finitely many indices carry each such h ; actual supports can only be smaller than the indicated sets. For the last assertion, each block support is contained in the union of the old supports and the blocks contributing at any exponent form a finite set, while coefficientwise associativity is finite. $\square$

Criterion (a) and criterion (b) are genuinely different tools and both are used below. Criterion (a) is the shorter route whenever the term valuations really do escape every bound. Criterion (b) is what survives at the boundary, where the term valuations increase strictly but remain trapped below a fixed element of Γ ; Theorem 8.1 is exactly such a case, and no leading-valuation argument alone can decide it.

Lemma 2.4 (Inward stability of strong evaluation). *Let $f = \sum_n a_n z^n \in K_\Gamma[[z]]$, and let $x, y \in K_\Gamma^\times$ with $v(y) \geq v(x)$. If $x \in \text{Dom}_{\text{str}}(f)$, then $y \in \text{Dom}_{\text{str}}(f)$. Thus membership of a nonzero argument in $\text{Dom}_{\text{str}}(f)$ depends only on its valuation, and entireness can be tested on monomial arguments.*

Proof. Put $b_n = a_n x^n$ and $u = y/x$, so (b_n) is strongly summable and $v(u) = \eta \geq 0$. Write $u = c t^\eta(1 + w)$, where $c \in \mathbb{C}^\times$ and w is zero or has positive valuation. Let S be $\text{supp}(w)$, with η adjoined when $\eta > 0$, and put $M = M(S)$. This is a well-ordered positive alphabet, so Lemma 2.1 makes M well ordered. Every finite power u^n has support in M .

Let $A = \bigcup_n \text{supp}(b_n)$. The supports of $a_n y^n = b_n u^n$ lie in $A + M$, a well-ordered set. For a fixed exponent γ there are only finitely many pairs $(\alpha, \mu) \in A \times M$ with $\alpha + \mu = \gamma$. For each such α , strong summability of (b_n) leaves only finitely many indices n . Hence γ occurs in only finitely many supports of $b_n u^n$, proving strong summability. Equal valuations allow the argument in both directions; compare an arbitrary nonzero y with $t^{v(y)}$ to obtain the monomial test. Evaluation at zero is always finite. $\square$

The statement fixes the full coefficient series a_n . It does not say that their leading valuations alone determine the domain: higher coefficient tails can still matter at a boundary scale.

2.2 Positive valuation is not cofinal decay

For $\beta > 0$ write

$$\Delta_\beta = \{\gamma \in \Gamma : |\gamma| \leq N\beta \text{ for some } N \in \mathbb{N}\}, \quad (2.1)$$

the smallest convex subgroup containing β . It is all of Γ exactly when β is an order unit. If it is proper, there is $\gamma > 0$ outside it, and then

$$\gamma > N\beta \quad \text{for every ordinary } N \in \mathbb{N}, \quad (2.2)$$

since a positive element bounded by some $N\beta$ would lie in Δ_β .

Remark 2.5 (Sufficiency is not necessity). Let $0 < \epsilon < H$ in Γ with $n\epsilon < H$ for every ordinary integer n ; for instance $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ ordered by the highest nonzero coordinate. The family $(t^{n\epsilon})_{n \geq 0}$ is strongly summable, since its support is an increasing sequence and each exponent occurs once, and its sum is the Hahn element $(1 - t^\epsilon)^{-1}$. Yet its valuations do not tend cofinally to $+\infty$. This is the most elementary warning against identifying positive valuation with topological nilpotence, and it is why criterion (b) of Lemma 2.3 is needed at all. No appeal to convergence of partial sums at the H scale is involved.

Remark 2.6 (The real-valued coarsening is imported, not reproved). When Γ does have an order unit h , the companion report [12, Lemma 9.1, `ent:lem:coarsening`] supplies an order-preserving $\pi : \Gamma \rightarrow \mathbb{R}$ with $\pi(h) = 1$ and kernel the convex subgroup $H = \{\gamma : n|\gamma| < h \text{ for all } n \geq 1\}$, hence a nontrivial real-valued non-Archimedean absolute value $|a|_\pi = \exp(-\pi(v(a)))$ inducing the intrinsic topology. We cite that lemma and do not reprove it. No proof in this report uses it: every argument below is carried out in Γ itself, which is what makes the arguments valid when no order unit exists. The coarsening is mentioned twice, as an optional rank-one picture of the order-unit case (Remark 8.3) and for the warning of Section 1.5 that the interpolation/Bézout dichotomy of that report and the annihilating-operator dichotomy of this one share an invariant but are different theorems.

2.3 The entire class and its closure properties

Polynomials are entire, because all the evaluation families are finite. An ordinary complex entire Taylor series need not be entire in the sense of Definition 1.2: at a monomial argument $x = t^\gamma$ the family

$$\left(\frac{x^n}{n!}\right)_{n \geq 0} \tag{2.3}$$

is strongly summable exactly when $\gamma > 0$, since $\gamma = 0$ puts infinitely many contributions at exponent zero and $\gamma < 0$ gives a descending sequence. Proposition 5.3 computes the exact domain of the corresponding family with an arbitrary Hahn argument.

Proposition 2.7 (Closure). $\mathcal{E}_\Gamma$ is a K_Γ -algebra, $\sigma_q(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$ for every $q \in K_\Gamma^\times$, and $D_z(\mathcal{E}_\Gamma) \subseteq \mathcal{E}_\Gamma$.

Proof. For addition, the evaluation supports lie in the union of two well-ordered sets with finite exponent multiplicities. For multiplication, the doubly indexed product family is strongly summable by Lemma 2.1 (1) and (3), and grouping by ordinary total degree preserves strong summability by the last clause of Lemma 2.3. Dilation is evaluation at qx . For $x \neq 0$ the derivative family is

$$(n+1)a_{n+1}x^n = x^{-1} \cdot (n+1)a_{n+1}x^{n+1}.$$

Discarding one term and multiplying each term by a nonzero ordinary integer enlarges neither supports nor multiplicities, and multiplication by x^{-1} preserves summability by Lemma 2.1 (3). At $x = 0$ only the constant term contributes. Iterate. $\square$

Proposition 2.8 (Faithfulness of evaluation). *If $f \in \mathcal{E}_\Gamma$ vanishes at infinitely many distinct ordinary complex arguments, then $f = 0$ as a formal series. Consequently a linear equation with rational-function coefficients that holds pointwise wherever its coefficients are defined is also a formal equation in the sense of Definition 1.4.*

Proof. Strong evaluation at 1 says that the coefficient family (a_n) has a well-ordered union of supports and is locally finite at every exponent. Hence for each γ

$$P_\gamma(T) = \sum_n (a_n)_\gamma T^n \in \mathbb{C}[T]$$

is a polynomial, and its value at an ordinary complex c is the coefficient of t^γ in $f(c)$. If f vanishes at infinitely many such c , every P_γ has infinitely many roots and is zero, so every a_n is zero. For the last assertion, clear denominators; the left-hand side is in $\mathcal{E}_\Gamma$ by Proposition 2.7, and nonzero polynomial denominators exclude at most finitely many ordinary complex arguments. $\square$

The nonexistence proofs may therefore be run on formal coefficients without weakening their reading as functional obstructions, and no interchange of infinite limits is needed anywhere.

The theorems below use only the valuation laws (1.1), the existence of every monomial t^γ , triviality of the valuation on $\mathbb{C}^\times$, and the support facts of this section. They do not require algebraic closedness, spherical completeness, Weierstrass preparation, or an embedding of Γ into $\mathbb{R}$.

2.4 Embedding the workspace into the surcomplex numbers

If Γ is an additive ordered subgroup of $\mathbf{No}$, the Conway normal-form map

$$\sum_{\gamma \in S} c_\gamma t^\gamma \mapsto \sum_{\gamma \in S} c_\gamma \omega^{-\gamma} \quad (2.4)$$

identifies the real-coefficient Hahn field with a subfield of $\mathbf{No}$ and its complexification with a subfield of $\mathbf{No}[i]$; a well-ordered S becomes a reverse-well-ordered set of Conway exponents. This viewpoint and its set-sized support discipline are standard [9, 10]. Here t^γ is a formal monomial and $\omega^{-\gamma}$ a Conway monomial; the notation does not silently identify every formal power with a choice of surreal exponentiation.

The results may therefore be read twice. Abstractly they concern Hahn fields and need no surreal construction. Concretely they restrict the exact linear descriptions available for surcomplex-valued power series on specified surreal scales. Formula (2.4) is the only bridge used, and no global surcomplex exponential is invoked.

3 The finite-step escape mechanism

The following isolates the mechanism used throughout. It needs no formula for the recurrence coefficients, no asymptotic slope for the valuations, and no assumption that cancellations are rare.

Lemma 3.1 (Escape chain). *Suppose a sequence (a_n) in K_Γ satisfies, for every $n \geq N$,*

$$c_0(n)a_n + \sum_{j=1}^s c_j(n)a_{n+j} = 0, \quad c_0(n) \neq 0, \quad (3.1)$$

and suppose $a_{n_0} \neq 0$ for some $n_0 \geq N$. Then $s \geq 1$ and there is an infinite sequence $n_0 < n_1 < n_2 < \dots$ with $j_r := n_{r+1} - n_r \in \{1, \dots, s\}$, with $a_{n_r} \neq 0$ and $c_{j_r}(n_r) \neq 0$ for every r , and with

$$v(a_{n_{r+1}}) - v(a_{n_r}) \leq v(c_0(n_r)) - v(c_{j_r}(n_r)). \quad (3.2)$$

Proof. Let $n \geq N$ with $a_n \neq 0$. The leftmost term of (3.1) is nonzero, so $s \geq 1$ and at least one later term is nonzero. If every nonzero later term had valuation strictly larger than $v(c_0(n)a_n)$, the leading monomial of the leftmost term could not cancel, contradicting (1.1). So some $j \in \{1, \dots, s\}$ has $c_j(n)a_{n+j} \neq 0$ and

$$v(c_j(n)a_{n+j}) \leq v(c_0(n)a_n),$$

which is (3.2). Taking the least eligible j makes the recursion definite. The successor coefficient is nonzero and its index is again at least N , so the construction never stops. $\square$

Remark 3.2 (The direction of the inequality matters). Solving the recurrence for its *last* coefficient typically yields a *lower* bound on a valuation, which obstructs nothing. The comparison above is made from the first nonzero coefficient forward and produces a later coefficient whose valuation is controlled from above. Cancellation among the other terms can raise the valuation of their sum, but cannot make every term on the right have valuation strictly larger than the single nonzero term on the left. Zeros and arbitrarily large cancellations elsewhere therefore do not invalidate the chain; the chain records only that finite inequality, repeatedly.

Theorem 3.3 (Evaluation exclusion criterion). *In the situation of Lemma 3.1, let $x \in K_\Gamma^\times$ and suppose that for every $n \geq N$ and every $j \in \{1, \dots, s\}$ with $c_j(n) \neq 0$,*

$$v(c_0(n)) - v(c_j(n)) + jv(x) \leq 0. \quad (3.3)$$

If $a_{n_0} \neq 0$ for some $n_0 \geq N$, then $(a_n x^n)_{n \geq 0}$ is not strongly summable. Equivalently: if $(a_n x^n)_{n \geq 0}$ is strongly summable, then $a_n = 0$ for every $n \geq N$.

Proof. Build the chain of Lemma 3.1 from n_0 and add $n_{r+1}v(x)$ to each side of (3.2), using $n_{r+1} = n_r + j_r$:

$$v(a_{n_{r+1}}x^{n_{r+1}}) \leq v(a_{n_r}x^{n_r}) + v(c_0(n_r)) - v(c_{j_r}(n_r)) + j_r v(x) \leq v(a_{n_r}x^{n_r}).$$

The evaluated leading exponents along the chain are therefore nonincreasing, and Lemma 2.2 applies. $\square$

Note that (3.3) is a weak inequality. This is the difference between an open and a closed excluded region, and it is available only because Lemma 2.2 also rules out a constant leading exponent. Section 5.2 exhibits an equation whose excluded region is exactly the closed one, and Section 8.5 exhibits a coarsened comparison where the boundary genuinely must be left in.

Corollary 3.4 (Uniform bounded cost; no divisibility). *Suppose in (3.1) there is a single $B \in \Gamma$ with $v(c_0(n)) - v(c_j(n)) \leq B$ for all $n \geq N$ and all $j \geq 1$ with $c_j(n) \neq 0$. Put $B^+ = \max(B, 0) \in \Gamma$. Then for every $x \neq 0$ with $v(x) \leq -B^+$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. Such x exist: $x = t^{-B^+}$, and $x = t^{-\delta}$ for every $\delta \geq B^+$.*

Proof. Since $B^+ \geq 0$ we have $v(x) \leq 0$, so $jv(x) \leq v(x) \leq -B^+ \leq -B$ for $j \geq 1$. Hence $v(c_0(n)) - v(c_j(n)) + jv(x) \leq B - B = 0$, and Theorem 3.3 applies. Every ordered abelian group admits a maximum of finitely many elements and the monomial t^{-B^+} exists, so no cofinal cyclic subgroup is needed. $\square$

The absence of a cofinality requirement here is exactly why the mechanism proves rigidity in groups of very large rank. It is also why the refined bound below, which divides by j , is a refinement of the constant and never of the conclusion.

Corollary 3.5 (Refined periodic threshold; requires Γ divisible). *Assume in addition that Γ is divisible. Assume each slot of (3.1) is either identically zero for $n \geq N$ or nonzero for all $n \geq N$ with periodic valuation there of common period $m \geq 1$; enlarge N past any initial nonperiodic range first. Let $\alpha_{j,r}$ be the valuation of slot j on the residue class $r \bmod m$, the identically zero slots being omitted. Put*

$$B_{\text{rec}} = \max_{\substack{0 \leq r < m \\ 1 \leq j \leq s, c_j \neq 0}} \frac{\alpha_{0,r} - \alpha_{j,r}}{j} \in \Gamma. \quad (3.4)$$

If there is no active forward slot $j \geq 1$, set $B_{\text{rec}} = 0$; the recurrence then forces $a_n = 0$ for all $n \geq N$ directly. Then for every $x \neq 0$ with $v(x) \leq -B_{\text{rec}}$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. In particular every solution with $a_n \neq 0$ for some $n \geq N$ — so in particular every nonpolynomial solution — satisfies

$$\text{Dom}_{\text{str}}(f) \subseteq \{0\} \cup \{x \in K_{\Gamma}^{\times} : v(x) > -B_{\text{rec}}\}, \quad f = \sum_n a_n z^n. \quad (3.5)$$

Proof. The case with no forward slot was handled in the statement. Otherwise divisibility makes each quotient $(\alpha_{0,r} - \alpha_{j,r})/j$ an element of Γ , so B_{rec} is defined. If $v(x) \leq -B_{\text{rec}}$ then $jv(x) \leq -j(\alpha_{0,r} - \alpha_{j,r})/j = -(\alpha_{0,r} - \alpha_{j,r})$, because multiplying a weak inequality by the positive integer j preserves it. Hence (3.3) holds and Theorem 3.3 applies. $\square$

The inequality in (3.5) is strict on the right: at $v(x) = -B_{\text{rec}}$ the chain may keep a constant leading exponent rather than descend, which is precisely why both halves of Definition 1.2 are needed. The bound is an *exclusion* bound. It does not assert that every point on the other side lies in the domain.

4 Valuations on integer and geometric orbits

4.1 Ordinary integers cannot create unbounded valuation loss

Lemma 4.1 (Integer evaluation). *Let $0 \neq P(X) = \sum_k p_k X^k \in K_{\Gamma}[X]$ and put $\beta(P) = \min_{p_k \neq 0} v(p_k)$. For all but finitely many nonnegative ordinary integers n ,*

$$P(n) \neq 0 \quad \text{and} \quad v(P(n)) = \beta(P). \quad (4.1)$$

Proof. The polynomial $t^{-\beta(P)}P$ has coefficients of nonnegative valuation and at least one of valuation zero, so its coefficientwise reduction

$$\overline{P}(X) = \sum_{v(p_k) = \beta(P)} \text{lc}(p_k) X^k \in \mathbb{C}[X]$$

is nonzero and has finitely many roots. The ordinary integers have distinct images in $\mathbb{C}$. At every integer n with $\overline{P}(n) \neq 0$, the Hahn coefficient of $t^{-\beta(P)}P(n)$ at exponent zero is nonzero, and no smaller exponent occurs. $\square$

Residue characteristic zero does real work here. The proof does not apply to p -adic integer inputs, where positive integers can have arbitrarily large valuation. Conversely no Diophantine approximation over $\mathbb{C}$ is relevant to v : every nonzero complex number, however small in ordinary modulus, has valuation zero.

4.2 A geometric orbit of nonzero valuation

Lemma 4.2 (Eventually affine valuation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(X) = \sum_k p_k X^k \in K_\Gamma[X]$. There are an index k_P with $p_{k_P} \neq 0$, the value $\beta_P = v(p_{k_P})$, and $N_P \in \mathbb{N}$, such that*

$$v(P(q^n)) = \beta_P + k_P n \lambda \quad (n \geq N_P). \quad (4.2)$$

In particular $P(q^n)$ is eventually nonzero.

Proof. The nonzero terms have valuations $v(p_k) + kn\lambda$. For two distinct indices the difference $v(p_k) - v(p_l) + (k - l)n\lambda$ is strictly monotone in n , so its sign is eventually constant and it vanishes for at most one integer, whether or not its values are cofinal in any convex subgroup. Finitely many pairs give one common threshold beyond which every pairwise comparison is strict and fixed, hence a unique minimizing index k_P . Its term cannot cancel against terms of strictly larger valuation. $\square$

Remark 4.3 (A rank-one shortcut that fails). The minimizing index need not be the smallest when $v(q) > 0$. In $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, take $q = t^\epsilon$ and $P(X) = X + t^{-H}$: the constant term has smaller valuation than q^n for every n , however large. Conversely with $P(X) = 1 + t^{-H}X$ the nonconstant term is the minimum for every n . The same phenomenon in the other sign convention: in $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, $q = t^{e_0}$ and $P(X) = t^{e_1} + X$ give $v(P(q^n)) = ne_0$ for every $n \geq 0$, so the nonzero constant coefficient never becomes the least-valued term. The proof above uses the eventual ordering of finitely many affine functions, not the Archimedean assertion that $n\epsilon$ eventually exceeds H , and not the rank-one slogan that q^n eventually beats every coefficient ratio.

4.3 Valuation-zero dilations, including residues that are roots of unity

Distinct powers of a valuation-zero q all have the same valuation, so the previous lemma says nothing. Polynomial evaluation nevertheless has only a finite periodic valuation pattern.

Theorem 4.4 (Unit-orbit valuations). *Let $q \in K_\Gamma^\times$ have $v(q) = 0$ and infinite multiplicative order, and let $0 \neq P \in K_\Gamma[X]$. Then $v(P(q^n))$ is eventually finite and periodic. More precisely, with $\zeta = \text{res}(q) \in \mathbb{C}^\times$:*

- (a) *if ζ is not a root of unity, $v(P(q^n))$ is eventually constant, equal to the minimum valuation of the coefficients of P ;*
- (b) *if ζ has exact order h , then $v(P(q^n))$ is eventually constant on each residue class modulo h , so its eventual period divides h .*

Proof. For (a) write $P(X) = \sum_k p_k X^k$ and $\beta = \min_k v(p_k)$. The coefficient of $P(q^n)$ at exponent β is $\overline{P}(\zeta^n)$ with $\overline{P}(X) = \sum_{v(p_k)=\beta} \text{lc}(p_k) X^k$, a nonzero complex polynomial. The points ζ^n are distinct, so this vanishes only finitely often, and no exponent below β occurs.

For (b) write

$$q = \zeta(1 + u), \quad \eta = v(u) > 0, \quad d = \text{lc}(u).$$

Here $u = q/\zeta - 1$ has positive valuation because q/ζ has valuation zero and residue 1, and $u \neq 0$ because otherwise $q = \zeta$ would have finite order. For every positive ordinary integer n the finite binomial expansion gives

$$y_n := (1 + u)^n - 1, \quad v(y_n) = \eta, \quad \text{lc}(y_n) = nd, \quad (4.3)$$

the leading coefficient being nonzero in characteristic zero. There is no convergence assertion: for each fixed n only finitely many powers occur.

Fix $r \in \{0, \dots, h-1\}$. For $n \equiv r \pmod{h}$ we have $\zeta^n = \zeta^r$, hence $q^n = \zeta^r(1 + y_n)$. Expand the nonzero polynomial

$$P(\zeta^r(1 + Y)) = \sum_{k=0}^{\deg P} b_{r,k} Y^k \quad (4.4)$$

and set

$$\beta_r = \min_{b_{r,k} \neq 0} (v(b_{r,k}) + k\eta), \quad J_r = \{k : v(b_{r,k}) + k\eta = \beta_r\}. \quad (4.5)$$

Then $P(q^n) = \sum_k b_{r,k} y_n^k$, and its coefficient at exponent β_r is the value at n of

$$R_r(T) = \sum_{k \in J_r} \text{lc}(b_{r,k}) d^k T^k \in \mathbb{C}[T]. \quad (4.6)$$

This polynomial is nonzero: its terms have distinct degrees and all the indicated coefficients are nonzero. Outside its finitely many integer roots the coefficient at β_r survives, so $v(P(q^n)) = \beta_r$ eventually on that class. Finitely many classes give one finite exceptional range. $\square$

Remark 4.5 (A second normalization of the same proof). Sources 08 and 09 both prove Theorem 4.4 (b), by two slightly different normalizations, and we record the other one because it is occasionally more convenient. Instead of factoring out the complex root of unity, put $q^h = 1 + u$ directly, with $\eta = v(u) > 0$ and $u \neq 0$ for the same reason. Write $n = r + hk$ and expand the polynomial $Q_r(Y) = P(q^r Y)$ as $Q_r(1 + V) = \sum_k b_{r,k} V^k$. The finite binomial expansion gives $V_k = (1 + u)^k - 1$ with $v(V_k) = \eta$ and $\text{lc}(V_k) = k \text{lc}(u)$, and $P(q^n) = Q_r(1 + V_k)$. The minimum construction (4.5) and the corresponding nonzero residue polynomial (4.6), now in the variable k rather than n , finish the argument. Neither version sums a binomial series to a noninteger exponent; every expansion is finite, and arbitrary supports in q and in the coefficients of P enter only through the leading valuation and leading coefficient of finitely many Hahn elements.

Example 4.6 (A genuine two-phase pattern). Let $q = -(1 + t^\eta)$ with $\eta > 0$. Then q is nontorsion, $\text{res}(q) = -1$ has order two, and for $P(X) = X - 1$ and $n \geq 1$

$$v(P(q^n)) = v(q^n - 1) = \begin{cases} 0, & n \text{ odd}, \\ \eta, & n \text{ even}. \end{cases}$$

For odd n the residue is -2 ; for even positive n the leading term is nt^η . The valuation sequence is periodic, not the dilation itself, and a claim of eventual *constancy* without splitting residue classes would be false. The escape argument uses the bounded valuation costs; it does not replace q by its residue.

Example 4.7 (Finite exceptional indices really occur). For $q = 1 + t^\eta$ and $P(X) = X - 1 - 7t^\eta$, the leading coefficient of $P(q^n)$ at exponent η is $n - 7$, so $v(P(q^n)) = \eta$ for positive $n \neq 7$ while $v(P(q^7)) = 2\eta$. With $P(X) = X - 1 - 3t^\eta$ the exceptional index is $n = 3$, where $P(q^3) = 3t^{2\eta} + t^{3\eta}$. The exceptions are harmless for the escape argument, since it needs only an eventual range, but they cannot be ignored in a statement asserted for all indices. This is why Lemma 4.1 and Theorem 4.4 both allow finitely many exceptions even when $\text{res}(q) = 1$.

For an ordinary complex q with $|q|_{\mathbb{C}} \neq 1$, the complex modulus does not alter these conclusions: as an element of the Hahn coefficient field, q still has valuation zero. The ordinary modulus and the scale valuation answer different questions.

5 Differential rigidity

5.1 From an operator to a forward recurrence

A formal series is D-finite exactly when its derivatives span a finite-dimensional space over $K_\Gamma(z)$. The D-finite/P-recursive equivalence is classical [1, Theorem 1.5]; the following records the exact finite conversion, because the valuations of the resulting recurrence, not ordinary coefficient growth, are what the proof consumes.

Lemma 5.1 (Differential coefficient recurrence). *Write*

$$L = \sum_{r=0}^R \sum_{k=0}^m c_{r,k} z^k D_z^r, \quad c_{r,k} \in K_\Gamma, \quad (5.1)$$

with at least one $c_{r,k} \neq 0$, and let $f = \sum_{n \geq 0} a_n z^n$. For $n \geq m$ the coefficient of z^n in Lf is

$$\sum_{s=-m}^R Q_s(n) a_{n+s}, \quad Q_s(X) = \sum_{r-k=s} c_{r,k} (X+s)_r, \quad (5.2)$$

where $(Y)_r = Y(Y-1)\cdots(Y-r+1)$ and $(Y)_0 = 1$. At least one Q_s is nonzero. Writing s_0 for the least active shift and $u = n + s_0$, every formal solution of $Lf = 0$ satisfies, for all sufficiently large u ,

$$\sum_{j=0}^s P_j(u) a_{u+j} = 0, \quad P_j(X) = Q_{s_0+j}(X - s_0), \quad P_0 \neq 0, \quad (5.3)$$

with $s = R - s_0$. If $s = 0$, every formal solution is a polynomial.

Proof. The operator $z^k D_z^r$ carries the term of index $n+r-k$ to degree n and multiplies it by $(n+s)_r$ with $s = r - k$, which gives (5.2). For a fixed shift s the distinct derivative orders r contribute polynomials of distinct degrees in X , so a nonempty group containing a nonzero coefficient cannot cancel identically and some Q_s is nonzero. Reindexing gives (5.3). If $s = 0$ the recurrence reads $P_0(u) a_u = 0$ with $P_0 \neq 0$, so by Lemma 4.1 all but finitely many a_u vanish. $\square$

Proposition 5.2 (Uniform obstruction for P-recursive coefficients). *Suppose (a_n) satisfies (5.3) for all $n \geq N_1$, with $P_j \in K_\Gamma[X]$ and $P_0 \neq 0$. Let $\beta_j = \beta(P_j)$ for the nonzero P_j and put*

$$B = \max(\{0\} \cup \{\beta_0 - \beta_j : j \geq 1, P_j \neq 0\}) \geq 0.$$

There is $N \geq N_1$, beyond the finitely many exceptional integers of Lemma 4.1 for the finitely many nonzero P_j , such that for every $x \neq 0$ with $v(x) \leq -B$, strong summability of $(a_n x^n)_n$ forces $a_n = 0$ for all $n \geq N$. If Γ is divisible one may replace B by the smaller threshold

$$B_{\text{rec}} = \max\left(\left\{\frac{\beta_0 - \beta_j}{j} : j \geq 1, P_j \neq 0\right\}\right), \quad (5.4)$$

which is (3.4) with period $m = 1$. If the displayed set is empty, take $B_{\text{rec}} = 0$; then every solution terminates beyond N without an evaluation hypothesis.

Proof. Discard the identically zero P_j . By Lemma 4.1 there is N beyond which $v(P_j(n)) = \beta_j$ for all the finitely many nonzero P_j simultaneously. Apply Corollary 3.4 with this B , or Corollary 3.5 with $m = 1$ when Γ is divisible. If no $j \geq 1$ remains, the recurrence forces the same vanishing directly. $\square$

These are certified exterior obstructions, not a computation of the exact evaluation domain of each solution. Individual solutions can have stricter restrictions or can terminate much earlier.

Proof of Theorem A. Lemma 5.1 converts L into (5.3). Take $B_L = B$ from Proposition 5.2 and take N_L to be an positive integer beyond the recurrence's starting index and beyond every nonnegative integer root of the finitely many residue polynomials $\overline{P_j}$. If a formal solution is strongly evaluable at one $x \neq 0$ with $v(x) \leq -B_L$, then $a_n = 0$ for all $n \geq N_L$, so its degree is less than N_L . An entire solution is in particular strongly evaluable at the available monomial t^{-B_L} , hence is a polynomial. Conversely a polynomial is entire and is annihilated by a sufficiently high derivative, which gives the displayed equality. $\square$

The closed region $v(x) \leq -B_L$, rather than an open one, comes from the weak inequality in Theorem 3.3, hence from Lemma 2.2. A polynomial solution of degree at least N_L would in any case contradict the recurrence at its last nonzero coefficient, since $P_0(n)$ no longer vanishes there.

Theorem A is stronger than merely excluding a globally extended exponential. The operator coefficients may themselves be arbitrary Hahn series carrying infinite surreal scales. There are nevertheless only finitely many of them, so their leading valuations give a finite obstruction that one additional exterior scale defeats.

5.2 An exact boundary, not only a nonexistence result

Proposition 5.3 (Formal exponential domain). *For $\eta \in \Gamma$ let*

$$E_\eta(z) = \sum_{n \geq 0} \frac{t^{n\eta}}{n!} z^n.$$

Then $D_z E_\eta = t^\eta E_\eta$ and

$$\text{Dom}_{\text{str}}(E_\eta) = \{0\} \cup \{x \in K_\Gamma^\times : v(x) > -\eta\}. \quad (5.5)$$

Proof. The differential identity is a formal coefficient comparison. Put $y = t^\eta x$ for $x \neq 0$, so the n th evaluation term is $y^n/n!$.

If $v(y) > 0$, write $y = ct^{v(y)}(1+u)$ with $c \in \mathbb{C}^\times$ and u zero or of positive valuation, and put $S = \text{supp}(u) \subseteq \Gamma_{>0}$. Since $(1+u)^n$ is a finite integer power, $\text{supp}(y^n) \subseteq h_n + M(S)$ with $h_n = nv(y)$, and (h_n) is strictly increasing, so its values are well ordered and each is taken once. Lemma 2.3 (b) makes the family strongly summable; the factor $1/n!$ is a nonzero complex scalar and changes no support.

If $v(y) < 0$ the leading exponents $nv(y)$ strictly decrease. If $v(y) = 0$, then $y^n/n!$ has the nonzero coefficient $\text{lc}(y)^n/n!$ at exponent zero for every n , so local finiteness fails. These cases exhaust $x \neq 0$, and $v(y) > 0$ is $v(x) > -\eta$. $\square$

The recurrence of E_η is $(n+1)a_{n+1} - t^\eta a_n = 0$, with one active shift $j = 1$, so $\alpha_0 = \eta$, $\alpha_1 = 0$, and (3.4) gives $B_{\text{rec}} = \eta$ with no division by an integer at all. Comparing with (5.5): the excluded region $v(x) \leq -\eta$ of Corollary 3.5 is here *exactly* the complement of the domain, boundary included. The general exclusion bound can therefore be attained, and the closed form of the boundary is not an artefact of a lossy estimate. This example is the reason for keeping Lemma 2.2 rather than the strictly descending version alone.

There is no conflict with any scalar exponential on a surreal field: E_η is an ordinary power series in z , not a total exponential evaluated by an independent construction. Taking $\eta = 0$ recovers the elementary observation (2.3).

5.3 Algebraic and system consequences

Corollary 5.4 (Algebraic entire series). *An element of $\mathcal{E}_\Gamma$ algebraic over $K_\Gamma(z)$ belongs to $K_\Gamma[z]$.*

Proof. Let $F = K_\Gamma(z, f)$ be the finite extension generated by the algebraic series. In characteristic zero its minimal polynomial is separable, and differentiating that polynomial expresses $D_z f$ inside F , because the derivative with respect to the algebraic variable is nonzero at f and hence invertible in F . So D_z preserves F , all derivatives of f lie in the finite-dimensional $K_\Gamma(z)$ -space F , and f is D-finite. Apply Theorem A. $\square$

This is not offered as a new algebraic rigidity theorem; it records that the linear result subsumes the algebraic case without any factorization theorem for $\mathcal{E}_\Gamma$.

Corollary 5.5 (Rational differential systems). *If a finite vector Y of strongly entire formal series satisfies $D_z Y = A(z)Y$ with $A \in M_r(K_\Gamma(z))$, every component of Y is a polynomial. The same holds for an inhomogeneous system with a rational forcing vector, provided all its solution components are strongly entire.*

Proof. The $K_\Gamma(z)$ -span of the components is stable under D_z , so the successive derivatives of each component span a finite-dimensional space. For $Y' = AY + b$ append the constant component 1 and use $\begin{pmatrix} A & b \\ 0 & 0 \end{pmatrix}$. $\square$

Proposition 5.6 (Rational inhomogeneous equations). *The polynomiality conclusions of this report remain valid when the right-hand side of a linear equation is a rational function of z , provided the corresponding homogeneous hypothesis on L or on q holds.*

Proof. Clear denominators in the coefficients and in the right-hand side. The result has a polynomial right-hand side, so the coefficient recurrence is homogeneous for all sufficiently large indices, and the escape statements were formulated for eventual recurrences. The left-hand operator is still nonzero, having been multiplied by a nonzero polynomial. $\square$

This applies, for instance, to the inhomogeneous first-order theta identity (8.2) when the parameter valuation is noncofinal: it is not necessary to homogenize it in order to see the obstruction.

Example 5.7 (A sparse recurrence does not escape the obstruction). The formal Airy-type equation $f'' - zf = 0$ has the recurrence

$$a_{n+3} = \frac{a_n}{(n+3)(n+2)} \quad (n \geq 0), \quad a_2 = 0.$$

With $a_0 = 1$ and $a_1 = 0$ there are infinitely many nonzero coefficients, at indices $3n$, all of valuation zero. At $x = t^{-\delta}$ their leading exponents are $-3n\delta$, strictly decreasing for every $\delta > 0$. Gaps between nonzero coefficients do not help: the escape chain follows only the surviving congruence class.

Remark 5.8 (What transcendence has actually been proved). Every nonpolynomial member of $\mathcal{E}_\Gamma$ is transcendental over $K_\Gamma(z)$ and satisfies no nonzero *linear* differential equation over $K_\Gamma(z)$, and no finite family $f, D_z f, \dots, D_z^R f$ is linearly dependent over $K_\Gamma(z)$. This does not imply differential transcendence in the nonlinear sense. A relation

$$P(z, f, D_z f, \dots, D_z^R f) = 0$$

involving products of f and its derivatives lies outside the present proofs unless it can separately be reduced to a finite linear system. The distinction is not cosmetic: the escape mechanism exploits a finite *linear* cancellation in which one nonzero coefficient forces a later nonzero coefficient, whereas nonlinear coefficient equations involve convolutions of many earlier coefficients and supply no such forward step without further work.

That further work is the nonlinear part of this report (Sections 14–17, from source 10). It does not repair the escape chain; it replaces the recurrence by a finite initial polynomial at one large scale. Every relation of order $R \leq 1$ is thereby excluded (Theorem D), and so is every relation of higher order whose residue corner polynomial is nonzero (Theorem 16.1). Relations of order at least two with a vanishing residue corner remain outside all proofs in this report.

6 Generic lines and several-variable rigidity

A single differential equation in several variables is not enough to force polynomiality: it could ignore most of the variables. The correct hypothesis is finite-dimensionality of the span of *all* mixed partial derivatives, which is what Theorem C assumes.

Lemma 6.1 (Avoiding countably many polynomial conditions). *For any countable family of nonzero polynomials in $K_\Gamma[z_1, \dots, z_d]$ there is $c \in \mathbb{C}^d$ at which none of them vanishes.*

Proof. For one polynomial H , take the least Hahn exponent among its finitely many nonzero coefficients; the coefficient at that exponent is a nonzero $\overline{H} \in \mathbb{C}[z_1, \dots, z_d]$, and $\overline{H}(c) \neq 0$ implies $H(c) \neq 0$. The coefficients of the countably many polynomials $\overline{H}$ form a countable subset of $\mathbb{C}$, and the field they generate over $\mathbb{Q}$ is countable. Choose $c_1, \dots, c_d$ algebraically independent over that field, successively outside the algebraic closure of a countable field, which remains countable while $\mathbb{C}$ is uncountable. $\square$

The same countable-field device is used in the companion report held in `docs/surcomplex/entire-functions-at-arbitrary-rank/` (`ent:mv:thm:exceptional`) to show that the set of polynomial directions of a nonpolynomial series is not all of projective space.

Proof of Theorem C. Suppose F is not a polynomial and write it by total degree,

$$F(z) = \sum_{n \geq 0} F_n(z), \quad F_n \text{ homogeneous of degree } n, \quad (6.1)$$

so infinitely many F_n are nonzero. Let $g_1, \dots, g_\sigma$ be a basis of the finite-dimensional $K_\Gamma(z_1, \dots, z_d)$ -span of all mixed derivatives of F , taken to be mixed derivatives themselves with $g_1 = F$. The span is stable under differentiation, so there are rational matrices A_i with

$$\partial_i g = A_i(z)g \quad (1 \leq i \leq d). \quad (6.2)$$

These matrices have finitely many nonzero denominator polynomials; choose one nonzero homogeneous component of each. Apply Lemma 6.1 to every nonzero F_n , to those finitely many chosen components, and to the coordinate polynomials z_i , obtaining $c \in \mathbb{C}^d$ with $F_n(c) \neq 0$ whenever $F_n \neq 0$ and with no denominator becoming the zero polynomial under $z = wc$. Then

$$h(w) = F(wc) = \sum_{n \geq 0} F_n(c)w^n$$

is nonpolynomial.

For each $x \in K_\Gamma$ the family $(a_\alpha c^\alpha x^{|\alpha|})_{\alpha \in \mathbb{N}^d}$ is strongly summable, by entireness of F at $(c_1x, \dots, c_dx)$. Group it into the finite blocks of fixed total degree; by the last clause of Lemma 2.3, $\sum_n F_n(c)x^n$ is strongly summable, so $h \in \mathcal{E}_\Gamma$.

The formal chain rule, applied after clearing denominators in (6.2), gives

$$\frac{d}{dw}g(wc) = \left(\sum_{i=1}^d c_i A_i(wc) \right) g(wc). \quad (6.3)$$

Every denominator here is a nonzero element of $K_\Gamma[w]$. It may vanish at $w = 0$, which is harmless: read the identity in the Laurent series field $K_\Gamma((w))$, with matrices in $M_\sigma(K_\Gamma(w))$. The finite $K_\Gamma(w)$ -span of the components of $g(wc)$ is preserved by d/dw , so its first component h is D-finite over $K_\Gamma(w)$, and Theorem A forces h to be a polynomial. This contradicts its construction. $\square$

Corollary 6.2 (Multivariate algebraic rigidity). *If a strongly entire $F \in K_\Gamma[[z_1, \dots, z_d]]$ is algebraic over $K_\Gamma(z_1, \dots, z_d)$, it is a polynomial.*

Proof. Each partial derivation extends to the finite separable extension generated by F , by differentiation of its minimal polynomial, so all mixed partial derivatives lie in one finite-dimensional space. Apply Theorem C. $\square$

The generic line is an existence device, not a promised effective choice of ordinary complex constants. The proof does not say that one prescribed line detects every nonpolynomial multivariate series; the line is chosen only after the countably many nonzero homogeneous parts and the finitely many rational connection matrices are known.

7 Dilation equations and the necessity of a cofinal scale

7.1 Exact coefficient conversion

Lemma 7.1 (Dilation coefficient recurrence). *Let $q \in K_\Gamma^\times$ have infinite multiplicative order and let $f = \sum a_n z^n$ satisfy (1.3). Write $p_\ell(z) = \sum_k p_{\ell,k} z^k$ and $Q_k(X) = \sum_\ell p_{\ell,k} X^\ell$, let d and e be the largest and smallest k with $Q_k \neq 0$, and put $s = d - e$. Then for every $n \geq 0$*

$$\sum_{h=0}^s R_h(q^n) a_{n+h} = 0, \quad R_h(X) = Q_{d-h}(q^h X), \quad (7.1)$$

with $R_0 = Q_d \neq 0$ and $R_s = Q_e(q^s \cdot) \neq 0$. Each nonzero R_h satisfies $R_h(q^n) \neq 0$ for all but finitely many n . If $s = 0$, every formal solution is a polynomial.

Proof. The coefficient of z^{n+d} in $\sum_{\ell} p_{\ell}(z)f(q^{\ell}z)$ is $\sum_{k=e}^d Q_k(q^{n+d-k})a_{n+d-k}$; substituting $h = d - k$ and using $Q_{d-h}(q^{n+h}) = Q_{d-h}(q^h \cdot q^n)$ gives (7.1). Since q has infinite order the points q^n are distinct, and a nonzero univariate polynomial vanishes at only finitely many of them; this already gives eventual nonvanishing, independently of the finer valuation results. If $s = 0$ the relation reads $R_0(q^n)a_n = 0$ for every n , so all but finitely many a_n vanish. $\square$

The powers q^h in (7.1) matter: dropping them changes the recurrence and can change its valuation constants. Equivalently, taking $m = \max_{\ell} \deg p_{\ell}$ and reading off the coefficient of z^{n+m} gives $\sum_j P_j(q^n)a_{n+j} = 0$ with $P_j(X) = \sum_{\ell} p_{\ell, m-j} q^{\ell j} X^{\ell}$ and $P_0 = Q_m \neq 0$; both conversions are checked on finite exact examples by the programs of Section 18. If $m = 0$ the equation already reads $P_0(q^n)a_n = 0$ and all solutions are polynomials with no entireness assumption at all, so we may assume $s \geq 1$ below.

7.2 The valuation-zero case

Corollary 7.2 (Every unit dilation is globally rigid). *Let $q \in K_{\Gamma}^{\times}$ have $v(q) = 0$ and infinite multiplicative order. Every linearly q -difference finite member of $\mathcal{E}_{\Gamma}$ is a polynomial, and each fixed equation (1.3) has a uniform exterior obstruction and a uniform degree bound. If Γ is in addition divisible, every nonpolynomial formal solution has its domain bounded as in (3.5) by the constant (3.4) formed from the eventual residue-class valuations of (7.1).*

Proof. Apply Theorem 4.4 simultaneously to the finitely many nonzero R_h of Lemma 7.1. Their valuations take only finitely many values after a common starting index, with common period one or the exact order h_0 of $\text{res}(q)$. All differences $v(R_0(q^n)) - v(R_h(q^n))$ are therefore bounded above by a single element of Γ , and Corollary 3.4 gives the exterior obstruction and, via the residue polynomials, the degree bound. When Γ is divisible, Corollary 3.5 applies with $m = 1$ if the residue has infinite order, and $m = h_0$ otherwise. In both cases take the starting index beyond the exceptional range. $\square$

This includes parameters arbitrarily close to a root of unity without being one. The finite exceptional cancellations of Theorem 4.4 can move the starting index but do not remove the bound.

7.3 Nonzero but noncofinal valuation

Proposition 7.3 (Noncofinal dilation scales force polynomiality). *Let $q \in K_{\Gamma}^{\times}$ with $\beta = v(q) \neq 0$ and suppose $|\beta|$ is not an order unit of Γ . Every linearly q -difference finite member of $\mathcal{E}_{\Gamma}$ is a polynomial, with a uniform exterior obstruction and a uniform degree bound for each fixed equation. Divisibility of Γ is not used.*

Proof. By Lemma 4.2 applied to the finitely many nonzero R_h of (7.1), there is a common index beyond which

$$v(R_h(q^n)) = \alpha_h + k_h n \beta, \quad k_h \in \mathbb{N}.$$

Put $V = |\beta| > 0$, choose $B \in \Gamma$ with $B \geq 0$ and $B \geq |\alpha_0 - \alpha_h|$ for all relevant h , and put $M = \max_h |k_0 - k_h| \in \mathbb{N}$. Then for $h \geq 1$

$$v(R_0(q^n)) - v(R_h(q^n)) = \alpha_0 - \alpha_h + (k_0 - k_h)n\beta \leq B + MnV. \quad (7.2)$$

Because $|\beta|$ is not an order unit, (2.2) supplies $\gamma > 0$ with $\gamma > NV$ for every $N \in \mathbb{N}$; in particular $MnV < \gamma$ for all n , trivially so if $M = 0$. Hence the right side of (7.2) is bounded above by the single element $B + \gamma$. Corollary 3.4 with this bound excludes every $x \neq 0$ with $v(x) \leq -(B + \gamma)$, and the explicit witness $x = t^{-(B+\gamma)}$ is available. Since an entire solution is evaluable there, it satisfies $a_n = 0$ beyond the common starting index, hence is a polynomial of degree less than that index. Nothing in this argument divides an element of Γ by an integer. $\square$

The proof covers two different higher-rank situations. The group may have no order unit at all, so every nonzero dilation valuation is noncofinal; alternatively the group may have an order unit while the chosen q lives on a smaller scale. Global q -difference finiteness depends on the valuation of that particular dilation, not only on the group. The argument also requires no asymptotic formula for $v(a_n)$ and no eventually quadratic coefficient valuation: a single infinite escape chain suffices, which is why cancellations and sparse zero coefficients do not force a separate Skolem–Mahler–Lech analysis.

7.4 An intrinsic coarsened exclusion bound

The previous proof produces more than one bad point. Let $\beta = v(q) \neq 0$, let $\Delta = \Delta_{|\beta|}$ be the convex subgroup (2.1), and assume $\Delta \neq \Gamma$. Let $\pi : \Gamma \rightarrow \Gamma/\Delta$ be the ordered quotient and put $w = \pi \circ v$. With the eventual constants α_h of the previous proof, put

$$\overline{B}_{\text{rec}}^{\flat} = \max(\{0\} \cup \{\pi(\alpha_0) - \pi(\alpha_h) : 1 \leq h \leq s, R_h \neq 0\}), \quad (7.3)$$

and, when Γ is divisible so that Γ/Δ is a divisible ordered group,

$$\overline{B}_{\text{rec}} = \max_{1 \leq h \leq s, R_h \neq 0} \frac{\pi(\alpha_0) - \pi(\alpha_h)}{h}. \quad (7.4)$$

Theorem 7.4 (Coarsened exclusion). *Let $f = \sum a_n z^n$ be a nonpolynomial formal solution of (1.3) with $v(q) \neq 0$ and $\Delta = \Delta_{|v(q)|}$ proper.*

- (a) *Under Convention 1.1 alone, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}^{\flat}$ lies outside $\text{Dom}_{\text{str}}(f)$.*
- (b) *If Γ is in addition divisible, every $x \neq 0$ with $-w(x) > \overline{B}_{\text{rec}}$ lies outside $\text{Dom}_{\text{str}}(f)$.*

Proof. Coarsening by Δ kills $n\beta$, since $n\beta \in \Delta$ for every n , so the projected valuations of the recurrence coefficients are the constants $\pi(\alpha_h)$. In case (b), $-w(x) > \overline{B}_{\text{rec}}$ gives $hw(x) < -(\pi(\alpha_0) - \pi(\alpha_h))$ for every relevant $h \geq 1$. In case (a) we have $w(x) < -\overline{B}_{\text{rec}}^{\flat} \leq 0$, hence $hw(x) \leq w(x) < -\overline{B}_{\text{rec}}^{\flat} \leq -(\pi(\alpha_0) - \pi(\alpha_h))$. Either way the projected leading valuation along an escape chain strictly decreases at every step. A strict decrease in the ordered quotient is a strict decrease in Γ , so the chain produces an infinite strictly descending sequence in the union of the supports. $\square$

The boundary is deliberately *not* excluded here. Equal leading values in the quotient can correspond to a strictly increasing sequence in Γ , and hence to a strongly summable family; Section 8.5 exhibits exactly that. Unlike Corollary 3.5, a coarsened comparison alone cannot decide local finiteness at an exponent of Γ .

8 Partial theta series: sharp witness and exact domain

8.1 The classical identities

For $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$ set

$$\Theta_p(z) = \sum_{n \geq 0} p^{\binom{n}{2}} z^n = \sum_{n \geq 0} p^{n(n-1)/2} z^n. \quad (8.1)$$

This is a standard partial theta series up to a harmless sign convention [5]. None of its coefficients vanishes, so it is not a polynomial. The coefficient identity $p^{n(n-1)/2} = p^{n-1} p^{(n-1)(n-2)/2}$ for $n \geq 1$ gives

$$\Theta_p(z) = 1 + z\Theta_p(pz), \quad (8.2)$$

and applying $\sigma_p - 1$ to $(1 - z\sigma_p)\Theta_p = 1$, with $\sigma_p z = pz\sigma_p$, yields the homogeneous equation

$$\Theta_p(z) - (1 + z)\Theta_p(pz) + pz\Theta_p(p^2z) = 0, \quad (8.3)$$

which is also obtained by substituting pz for z in (8.2) and subtracting the two inhomogeneous identities. The existence of these identities is not the new part. What matters is precisely where the series is strongly evaluable, for an arbitrary-rank value group and an arbitrary Hahn parameter.

8.2 Exact domain with arbitrary coefficient tails

Theorem 8.1 (Exact partial theta domain). *Let Γ be as in Convention 1.1 and let $p \in K_\Gamma^\times$ with $\beta = v(p) > 0$, and let Δ_β be as in (2.1). Then*

$$\begin{aligned} \text{Dom}_{\text{str}}(\Theta_p) &= \{0\} \cup \{x \in K_\Gamma^\times : v(x) \geq -N\beta \text{ for some } N \in \mathbb{N}\} \\ &= \{0\} \cup \{x \in K_\Gamma^\times : (v \bmod \Delta_\beta)(x) \geq 0\}. \end{aligned} \quad (8.4)$$

In particular $\Theta_p \in \mathcal{E}_\Gamma$ if and only if β is an order unit of Γ . No monomial hypothesis on p or on x is needed, and Γ need not be divisible.

Proof. Evaluation at 0 is finite. For $x \neq 0$ put $\delta = v(x)$ and write

$$p = ct^\beta(1 + u), \quad x = dt^\delta(1 + w), \quad c, d \in \mathbb{C}^\times,$$

where each of u, w is zero or has strictly positive valuation. Let $S = \text{supp}(u) \cup \text{supp}(w) \subseteq \Gamma_{>0}$ and $M = M(S)$. The n th evaluation term is

$$p^{\binom{n}{2}} x^n = c^{\binom{n}{2}} d^n t^{h_n} (1 + u)^{\binom{n}{2}} (1 + w)^n, \quad h_n = \binom{n}{2} \beta + n\delta,$$

and the last two factors are finite integer powers, so

$$\text{supp}(p^{\binom{n}{2}} x^n) \subseteq h_n + M, \quad v(p^{\binom{n}{2}} x^n) = h_n. \quad (8.5)$$

Suppose first that $\delta \geq -N\beta$ for some $N \in \mathbb{N}$. Then

$$h_{n+1} - h_n = n\beta + \delta \geq (n - N)\beta > 0 \quad (n > N),$$

so (h_n) is eventually strictly increasing. Its set of values is therefore well ordered, including the finite initial segment, and each value is taken by finitely many indices. Lemma 2.3(b) applied to (8.5) proves strong summability.

If no such N exists, then $\delta < -N\beta$ for every N , so $h_{n+1} - h_n = n\beta + \delta < 0$ for every n . By (8.5) the leading exponents are exactly h_n , and they form a strictly decreasing sequence, so strong summability fails by Lemma 2.2.

Finally, $\delta \geq -N\beta$ for some N says exactly that the image of δ in Γ/Δ_β is nonnegative: a nonnegative coset either is positive, whence $\delta > 0$, or is zero, whence $\delta \in \Delta_\beta$ and $\delta \geq -N\beta$ for some N ; a negative coset lies below every element of Δ_β . The last assertion is the case $\Delta_\beta = \Gamma$. $\square$

This proof controls the whole support, not only the leading valuation. Leading valuation formulas by themselves do not in general determine a Hahn evaluation domain, because hidden tails can matter; here the common positive monoid M is the extra certificate that makes the formula exact. Note also that the entire calculation is carried out in Γ : nothing is divided by an integer, and no real-valued surrogate for v is introduced.

Remark 8.2 (A second, shorter proof of the order-unit half). When $\mu = \beta$ is an order unit, entireness also follows from the cofinal-valuation criterion Lemma 2.3(a) alone, and the routes of sources 08 and 09 are both kept here because they are genuinely different. For $x \neq 0$ with $\alpha = v(x)$, the n th evaluated term has valuation

$$\frac{n(n-1)}{2}\mu + n\alpha. \quad (8.6)$$

Given a target $\beta' \in \Gamma$, choose ordinary nonnegative integers M', B' with $\alpha \geq -M'\mu$ and $\beta' \leq B'\mu$, which is possible exactly because μ is an order unit. Then (8.6) is at least $(\frac{n(n-1)}{2} - M'n)\mu$, which exceeds $B'\mu$ for all sufficiently large ordinary n . So the term valuations tend cofinally to $+\infty$ and Lemma 2.3(a) applies. Only leading term valuations are used, so unrestricted higher Hahn tails in p do not invalidate the conclusion. This shorter route is available only in the order-unit case: at the boundary $v(x) = -N\beta$ with β noncofinal, the term valuations increase strictly but stay trapped below a fixed element of Γ , so criterion (a) says nothing and the monoid certificate is essential.

Remark 8.3 (The order-unit case has an optional rank-one picture). When β is an order unit, the real-valued coarsening $\pi : \Gamma \rightarrow \mathbb{R}$ of [12, Lemma 9.1], normalized by $\pi(\beta) = 1$, gives the same field K_Γ a rank-one absolute value. For $x \neq 0$, its theta terms have coarsened valuations $\binom{n}{2} + n\pi(v(x))$, tending to $+\infty$. This is an optional normed-field picture of the order-unit case, not a map that replaces the Hahn exponents by their images in $\mathbb{R}$.

Such a coefficientwise replacement need not define a Hahn series. In $\mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating all $n\epsilon$, the coarsening normalized by H kills ϵ . The valid series $\sum_{n \geq 0} t^{n\epsilon}$ would then place infinitely many coefficients at exponent zero. Coarsening a valuation retains the field and changes the value map; it is not an embedding into a complex-coefficient Hahn field with smaller exponent group. No proof here requires this optional coarsening.

8.3 Completion of the sharp criterion

Proof of Theorem B. Assume (i) and suppose (ii) fails. If $v(q) = 0$, Corollary 7.2 makes every entire solution a polynomial. If $v(q) \neq 0$ and $|v(q)|$ is not an order unit, Proposition 7.3 does the same. Either way (i) fails, which proves (i) $\Rightarrow$ (ii) and supplies the stated uniform obstruction and degree bound.

Assume (ii). If $v(q) > 0$ is an order unit take $p = q$: by Theorem 8.1, $\Theta_q \in \mathcal{E}_\Gamma \setminus K_\Gamma[z]$, and (8.3) is a nonzero linear q -difference equation with polynomial coefficients. If $v(q) < 0$ and $-v(q)$ is an order unit take $p = q^{-1}$, so $v(p) = -v(q) > 0$ is an order unit and $\Theta_{q^{-1}} \in \mathcal{E}_\Gamma$ by the same theorem. Substituting q^2z for z in (8.3), with $p = q^{-1}$ so that $pz \mapsto qz$ and $p^2z \mapsto z$, gives

$$\Theta_{q^{-1}}(q^2z) - (1 + q^2z)\Theta_{q^{-1}}(qz) + qz\Theta_{q^{-1}}(z) = 0, \quad (8.7)$$

which uses only nonnegative powers of σ_q and has polynomial coefficients that are not all zero. Both signs of $v(q)$ are covered. $\square$

The partial theta function is used as an established type of construction, not advertised as a newly discovered special function. Its role is precise: it realizes the positive side of a necessary and sufficient criterion, including nonmonomial q and arbitrary valuation rank.

8.4 Two scales in the same rank-two field

Example 8.4 (One field, opposite answers). Let $\Gamma = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ with $e_1 \gg e_0 > 0$, ordered by the highest nonzero coordinate. Then e_1 is an order unit and e_0 is not.

For $q = t^{e_0}$, no nonpolynomial strongly entire series satisfies a nonzero linear q -difference equation. The series Θ_q itself is not entire: by Theorem 8.1 its exact domain is the valuation ring obtained by forgetting the e_0 coordinate, which contains every x with $v(x) = -Ne_0$ and excludes $x = t^{-e_1}$.

For $q = t^{e_1}$, Θ_q is strongly entire, though the value group still has rank two. The two conclusions occur in the *same* Hahn field: the distinction is the dilation scale, not the rank of the field.

Under $e_0 = 1$, $e_1 = \omega$ and $t^\gamma = \omega^{-\gamma}$, the first dilation is multiplication by ω^{-1} and the second by $\omega^{-\omega}$. Quadratic multiples of the first valuation never reach the second scale; quadratic multiples of the second dominate every scale in this workspace. A fuller table for the same group, including unit dilations and the torsion exception, appears in Section 11.3.

8.5 The coarse boundary really can be included

For the theta series the forward recurrence is $a_{n+1} = p^n a_n$, read off from (8.2), whose inhomogeneity affects only the index $n = 0$ and is removed by Proposition 5.6. Its two slots have valuations $n\beta$ and 0, both of which project to 0 modulo Δ_β , so the projected gap is zero and $\overline{B}_{\text{rec}} = \overline{B}_{\text{rec}}^\flat = 0$. Theorem 7.4 therefore excludes the points of negative coarsened valuation, while Theorem 8.1 includes *all* points of zero coarsened valuation. The strict inequality in the coarsened exclusion theorem cannot be replaced by a weak one in general. This is the exact mirror image of Section 5.2, where the uncoarsened bound was attained with its boundary excluded from the domain. The two boundaries behave differently because one comparison is made in Γ and the other in a quotient that has forgotten the scale doing the work.

9 Torsion parameters and why they are excluded

If $q^h = 1$ then $\sigma_q^h - 1$ annihilates *every* formal power series, so the unrestricted definition of linear q -difference finiteness is vacuous for such a parameter. This is not a technical inconvenience removable by a better proof; it is a genuine counterexample to dropping the nontorsion hypothesis in Theorem B.

The difficulty persists even after excluding operators with zero action. For $h > 1$ and

$$f(z) = z^r g(z^h),$$

one has $f(qz) = q^r f(z)$, and the operator $\sigma_q - q^r$ is not zero on all formal series when $h > 1$. A nonpolynomial entire g produces a nonpolynomial entire f , since evaluation of f at x is evaluation of g at x^h followed by multiplication by x^r . Finite-order dilations therefore have a real periodic-support exception and are not a limiting case of Theorem B. In particular “the residue of q is a root of unity” and “ q is a root of unity” are fundamentally different hypotheses: Example 4.6 is covered by the theorem and this section is not.

There is nevertheless a meaningful eigenspace statement.

Proposition 9.1 (Torsion covariance). *Let $q \in \mathbb{C}^\times$ have exact finite order m and let $0 \leq r < m$.*

- (a) *A formal series satisfies $f(qz) = q^r f(z)$ if and only if $f(z) = z^r G(z^m)$ for a unique $G \in K_\Gamma[[z]]$.*
- (b) *If $G \in \mathcal{E}_\Gamma$ then $f = z^r G(z^m) \in \mathcal{E}_\Gamma$.*
- (c) *The converse of (b) also holds, so $f \in \mathcal{E}_\Gamma$ if and only if $G \in \mathcal{E}_\Gamma$, without divisibility of Γ .*

Moreover, every root of unity of K_Γ lies in $\mathbb{C}$, so the hypothesis $q \in \mathbb{C}^\times$ loses no torsion element of the Hahn field. All parts use only Convention 1.1.

Proof. For (a), coefficient comparison gives $(q^n - q^r)a_n = 0$, so only degrees congruent to r modulo m occur, and G is read off uniquely.

For (b), evaluation of $z^r G(z^m)$ at x is x^r times the family for G at x^m , reindexed; multiplication by the fixed element x^r preserves strong summability by Lemma 2.1 (3), and at $x = 0$ the evaluation is finite.

For (c), suppose $f \in \mathcal{E}_\Gamma$ and let $y \in K_\Gamma^\times$. Set $\delta = v(y)$, $\gamma = \min(\delta, 0)$, and $x = t^\gamma$. Since $m \geq 1$, we have $v(x^m) = m\gamma \leq \delta$. Strong evaluation of f at x , restricted to the degrees $r + mn$ and multiplied by x^{-r} , gives strong evaluation of G at x^m . Lemma 2.4 then gives strong evaluation at y . At $y = 0$ evaluation is finite. No root of y is extracted.

Finally, let $\zeta \in K_\Gamma$ be a root of unity. Then $v(\zeta) = 0$; dividing by $\text{res}(\zeta)$ gives $1 + u$ with u zero or of positive valuation, and $(1 + u)^{m'} = 1$ for some $m' \geq 1$. If $u \neq 0$ then $(1 + u)^{m'} - 1$ has valuation $v(u)$ and leading coefficient $m' \text{lc}(u) \neq 0$ in characteristic zero, a contradiction. So $u = 0$ and $\zeta = \text{res}(\zeta) \in \mathbb{C}$. $\square$

The converse uses only that the values $m\gamma$ are coinital in Γ , as the explicit choice $\gamma = \min(\delta, 0)$ shows. Actual m th roots need not exist in the original Hahn field; this stronger algebraic requirement is unnecessary for the entireness equivalence.

10 An extension to mixed differential–dilation equations

For a nonzero noncofinal dilation valuation, derivatives do not repair the missing-scale problem. The extension is useful when a symbolic system allows both differentiation and rescaling of the argument.

Lemma 10.1 (Polynomial dependence on the index as well as the dilation). *Let $q \in K_\Gamma^\times$ with $\lambda = v(q) \neq 0$ and let $0 \neq P(U, X) \in K_\Gamma[U, X]$. There are $\beta \in \Gamma$, $k \in \mathbb{N}$ and $N \in \mathbb{N}$ with*

$$v(P(n, q^n)) = \beta + kn\lambda \quad (n \geq N).$$

Proof. Write $P(U, X) = \sum_k C_k(U)X^k$. By Lemma 4.1 each nonzero $C_k(n)$ has the fixed valuation $\beta_k = \beta(C_k)$ for all sufficiently large n . The finitely many summands then have valuations $\beta_k + kn\lambda$, and the pairwise-order argument of Lemma 4.2 produces a unique eventual minimizing index, whose term cannot cancel. $\square$

Theorem 10.2 (Mixed rigidity at a noncofinal nonzero scale). *Let Γ be as in Convention 1.1, let $q \in K_\Gamma^\times$ with $v(q) \neq 0$, and suppose $|v(q)|$ is not an order unit of Γ . If a strongly entire f satisfies a nonzero equation*

$$\sum_{r,\ell} p_{r,\ell}(z) D_z^r \sigma_q^\ell f(z) = 0, \quad p_{r,\ell} \in K_\Gamma[z], \quad (10.1)$$

where the sum is finite and the coefficients are not all zero, then f is a polynomial. Each fixed equation has a uniform exterior obstruction and a uniform degree bound.

Proof. Expand $p_{r,\ell}(z) = \sum_k c_{r,\ell,k} z^k$. With $s = r - k$, the term of coefficient index $n + s$ contributes to degree n the quantity $c_{r,\ell,k}(n + s) \underline{r} q^{\ell(n+s)} a_{n+s}$, so the coefficient recurrence has shift polynomials

$$P_s(U, X) = \sum_{r-k=s} \sum_{\ell} c_{r,\ell,k} q^{\ell s} (U + s) \underline{r} X^\ell, \quad (10.2)$$

evaluated at $(U, X) = (n, q^n)$. For fixed s and ℓ , distinct r give distinct degrees in U , so a nonempty group of nonzero coefficients cannot cancel identically and some P_s is nonzero. Reindex from the smallest active shift; replacing U by $U - s_0$ and X by $q^{-s_0} X$ preserves polynomiality and nonzeroness, so the leftmost coefficient is $A_0(n) = P'_0(n, q^n)$ for a nonzero bivariate P'_0 , and Lemma 10.1 makes it eventually nonzero.

Lemma 10.1 gives eventual affine valuations for all the nonzero coefficients, so their differences obey a bound $B + Mn|v(q)|$ exactly as in (7.2). Since $|v(q)|$ is not an order unit, (2.2) bounds this by one element of Γ , and Corollary 3.4 finishes the proof. $\square$

The hypothesis $v(q) \neq 0$ is essential to *this* proof. For $v(q) = 0$ the separate pure differential and pure dilation conclusions have been proved above, but they are not a proof of the mixed conclusion, and no mixed unit-valuation classification is asserted anywhere in this report; see Question 19.4.

11 Surreal workspaces that separate the theories

The results so far are stated in set-sized Hahn fields, which keeps the foundations transparent and allows a literal surreal reading without quantifying over a proper class as though it were a set.

11.1 The workspace without an order unit

Take

$$\Gamma_\infty = \bigoplus_{j \geq 0} \mathbb{Q} e_j, \quad 0 < e_0 \ll e_1 \ll e_2 \ll \cdots, \quad (11.1)$$

ordered so that the highest index with a nonzero coordinate determines the sign, equivalently $e_{j+1} > N e_j > 0$ for all $j \geq 0$ and $N \geq 1$. Every element has only finitely many nonzero coordinates. One may identify e_j with the surreal ω^j .

The group has no order unit: if $\gamma > 0$ has largest active index j , then $e_{j+1} > n\gamma$ for every ordinary n . The sequence (e_n) is nevertheless cofinal in the whole group, so this is a countable

cofinality that cannot be generated by repeatedly adding one positive element. This group and this distinction appear in the companion entire-function report [12] and are background here, not a new observation.

Corollary 11.1 (Universal nontorsion dilation rigidity). *Suppose Γ has no order unit. Then every nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfies, simultaneously for all nontorsion $q \in K_\Gamma^\times$ and all $r \geq 0$,*

$$f(z), f(qz), \dots, f(q^r z) \quad \text{are linearly independent over } K_\Gamma(z).$$

Such fields can contain nonpolynomial strongly entire series, as (11.1) and Theorem 11.2 show.

Proof. A linear dependence over $K_\Gamma(z)$ becomes, after clearing denominators, a nonzero equation (1.3). Since Γ has no order unit, no $|v(q)|$ is an order unit, so Theorem B forces f to be a polynomial. $\square$

Theorem 11.2 (An explicit universally nonholonomic entire series). *In K_{Γ_∞} the series*

$$F(z) = \sum_{n \geq 0} t^{e_n} z^n = \sum_{n \geq 0} \omega^{-\omega^n} z^n \tag{11.2}$$

is strongly entire and nonpolynomial. It satisfies no nonzero linear differential equation over $K_{\Gamma_\infty}(z)$ and, for every nontorsion $q \in K_{\Gamma_\infty}^\times$, no nonzero polynomial-coefficient linear q -difference equation. For every q of nonzero valuation it satisfies no nonzero mixed equation (10.1) either.

Proof. Fix $x \neq 0$ and write $x = ct^\delta(1+w)$ with $c \in \mathbb{C}^\times$ and w zero or of positive valuation. Choose M with δ involving only the coordinates $e_0, \dots, e_M$. The n th evaluation term has leading exponent $h_n = e_n + n\delta$ and support contained in $h_n + M(\text{supp } w)$. For $n > M$ the highest coordinate of h_n is e_n with coefficient 1, so $h_n > 0$, and the highest coordinate of

$$h_{n+1} - h_n = e_{n+1} - e_n + \delta$$

is e_{n+1} with coefficient 1, so (h_n) is eventually strictly increasing. Lemma 2.3 (b) proves strong summability; at $x = 0$ only the constant term remains. All coefficients are nonzero, so F is nonpolynomial.

Alternatively, for any $\beta \in \Gamma_\infty$ choose j larger than every active coordinate index of δ and of β ; for $n > j$ the largest active coordinate of $h_n - \beta$ is n with coefficient 1, so $h_n > \beta$, and Lemma 2.3 (a) applies for arbitrary w . Cofinal leading valuations control the whole support, since every exponent of a Hahn term is at least its leading exponent.

The differential conclusion is Theorem A. Since Γ_∞ has no order unit, Theorem B — equivalently Corollary 11.1 — excludes every nontorsion dilation equation, and Theorem 10.2 excludes the mixed equations at every q with $v(q) \neq 0$. Rational coefficient equations reduce to the polynomial-coefficient case by clearing denominators. $\square$

The universal quantifier over q matters. The parameter may be chosen after F is fixed and may itself have infinite support; each individual $v(q)$ still has finitely many coordinates in Γ_∞ , hence is noncofinal. So the theorem excludes a linear relation for *each* such parameter, not merely for an advance list of simple dilations. The series has a simple explicit coefficient rule, and this is a statement about finite linear annihilating equations, not a claim that the rule is uncomputable or lacks a finite description.

11.2 Why the same formula is not entire on all surcomplex numbers

Let $e_* = \omega^\omega$, which dominates every integer multiple of every positive element of Γ_∞ , and enlarge the group to $\Gamma_\infty + \mathbb{Q}e_*$. At the surcomplex point $x = \omega^{e_*} = t^{-e_*}$ the evaluated leading exponents are $e_n - ne_*$, whose successive differences $e_{n+1} - e_n - e_*$ are negative. The defining family is therefore not strongly summable, by Lemma 2.2.

This explicit obstruction prevents the fixed-workspace theorem from being misread as an assertion of a nonpolynomial power series entire on all of $\mathbf{No}[i]$. The example illustrates two independent restrictions: a workspace can be large enough that no single dilation scale is cofinal, yet small enough to admit a countable sequence of coefficient scales escaping all its arguments, and enlarging it further can destroy the very entirety on which the theorem operates. The companion report studies this kind of boundary in general [12]; the calculation here is included only to state the domain of the present example unambiguously, and no new universal scalar-extension theorem is claimed.

11.3 Two classification tables

Let $\Gamma = \mathbb{Q}H \oplus \mathbb{Q}\epsilon$ with H dominating every $n\epsilon$, so that H is an order unit and ϵ is not.

Dilation q	Its valuation	Nonpolynomial entire solution of a nonzero linear q -difference equation?
t^H	H	Yes: Θ_{t^H} , by (8.3).
t^{-H}	$-H$	Yes: Θ_{t^H} , by (8.7).
t^ϵ	ϵ	No: positive, but noncofinal.
$1 + t^H$	0	No: an infinite-order unit.
2	0	No: complex size does not change the Hahn valuation.
-1	0	Outside the theorem: a torsion parameter, with periodic-support exceptions.

Workspace group	Parameter	Solution exists?	Reason
$\mathbb{Q}$	$v(q) \neq 0$	Yes	Every positive nonzero scale is an order unit.
$\mathbb{Q}$	$v(q) = 0$, nontorsion	No	Unit-orbit valuations are eventually periodic.
$\mathbb{Q}e_0 \oplus \mathbb{Q}e_1$, $e_1 \gg e_0$	$v(q) = e_0$	No	A lower scale is not cofinal.
the same group	$v(q) = e_1$	Yes	The top scale is an order unit.
$\bigoplus_{n \geq 0} \mathbb{Q}e_n$	any nontorsion q	No	No scale is an order unit, although F is entire.

In every row, nonpolynomial strongly entire D-finite series are impossible. Torsion parameters are deliberately absent from the second table: their annihilating operator may be an identity on every series. Finite valuation rank alone is therefore not the criterion, and the distinction persists for nonmonomial dilations, because the theorem uses the valuation rather than a requirement that q be a monomial.

Corollary 11.3 (Detecting order units by functional equations). *For a nonzero set-sized ordered abelian group Γ the following are equivalent:*

- (i) Γ has an order unit;
- (ii) some nontorsion $q \in K_\Gamma^\times$ admits a nonpolynomial strongly entire solution of a nonzero polynomial-coefficient linear q -difference equation.

Proof. Theorem B gives (ii) $\Rightarrow$ (i). For the converse take $q = t^\mu$ for an order unit μ ; then $v(q) = \mu$ is an order unit and q is nontorsion. $\square$

This recovers a property of the ordered group from a global functional equation existence question. The workspace (11.1) shows why the existence of nonpolynomial entire functions is by itself strictly weaker than the existence of an order unit.

11.4 Several meanings of transcendence, kept apart

For the explicit F the established conclusions are: transcendence over $K_{\Gamma_\infty}(z)$ by Corollary 5.4; linear differential independence of every finite family of derivatives; and linear independence along every single nontorsion geometric dilation orbit. These are substantial restrictions, and they do *not* imply that no polynomial relation $P(z, F, D_z F, \dots, D_z^R F) = 0$ exists. Nonlinear differential equations lie outside these proofs, and a relation involving $F(q_1 z)$ and $F(q_2 z)$ for multiplicatively independent parameters is not covered merely by applying the single- q theorem twice.

The nonlinear part adds one more conclusion by a separate proof: F and $D_z F$ are algebraically independent over $K_{\Gamma_\infty}(z)$ (Corollary 17.2), so every nonlinear relation with $R \leq 1$ is excluded. Relations with $R \geq 2$ are still not excluded unless their residue corner polynomial is nonzero, and no statement about several dilations follows.

12 Coefficient fields, degenerate value groups, characteristic

The one-variable proofs work over any characteristic-zero coefficient field with trivial valuation: the root-of-unity argument uses only finitely many ordinary polynomial roots and needs no algebraic closedness, and the integer-evaluation lemma needs only residue characteristic zero. All theorems also hold over $\mathbb{R}((t^\Gamma))$, the generic-line proof of Theorem C using uncountability of $\mathbb{R}$ in place of uncountability of $\mathbb{C}$. The results therefore concern surreal as well as surcomplex workspaces. More generally the core versions work over $k((t^\Gamma))$ for any characteristic-zero field k , and, as Section 1.4 records, without divisibility of Γ .

No statement here is a theorem for arbitrary valued fields of positive residue characteristic. In that setting Lemma 4.1 fails and the valuations of near-torsion geometric powers can behave differently, so any comparison with the ultrametric literature must keep the coefficient field and the residue characteristic visible. Example 17.3, from source 10, shows that characteristic zero is needed even for the single linear equation $D_z f = 0$: over $\mathbb{F}_p((t^\mathbb{Q}))$ a nonpolynomial strongly entire series has zero derivative.

The nonlinear part (Convention 14.1) is proved by its source over an arbitrary field of characteristic zero. That is a statement about Sections 14–17 only; the generality of the linear theorems is governed by the first paragraph of this section and its own argument.

We assumed $\Gamma \neq 0$ in order to speak of positive exterior scales. When $\Gamma = 0$ the field is $\mathbb{C}$ with the trivial valuation, strong evaluation at $x = 1$ already permits only finitely many nonzero coefficients because every nonzero term has support $\{0\}$, and entire series are polynomials for that immediate support reason.

13 Quantitative certificates and symbolic representation

13.1 What one can extract from an annihilating equation

The classification has a direct representational consequence. A system whose nonpolynomial special functions are represented only by finite linear differential equations cannot cover the nonpolynomial strong entire functions of a Hahn workspace. Allowing linear dilation equations enlarges the class only when a cofinal dilation scale is available. In the no-order-unit example, an exact coefficient rule such as (11.2) lies outside both pure equation formats.

This does not rule out exact symbolic computation with such functions. It specifies what a faithful representation must keep separate: the coefficient-generation rule, the exponent group, the strong evaluation domain, and the type of finite equation being asserted. A field-independent label such as “entire theta function” discards relevant information; for Θ_p the proof certificate is not merely $v(p) > 0$, it includes that $v(p)$ is an order unit of the selected group.

For a differential operator the reduction yields a concrete certificate. Compute the finite shift polynomials Q_s of (5.2) and shift the least active index to zero. For each resulting P_j record its least coefficient valuation β_j and the corresponding nonzero complex residue polynomial $\overline{P_j}$. Record an integer N beyond their nonnegative integer roots and beyond the recurrence’s starting index. Record the threshold B of Proposition 5.2, or (5.4) when Γ is divisible. The escape statements then certify that every formal solution strongly evaluable at some $x \neq 0$ with $v(x) \leq -B$ has degree less than N .

For a valuation-zero dilation the same plan uses (7.1): if $\text{res}(q)$ has infinite order use the minimum coefficient valuation, and if it has order h use (4.4)–(4.6) on each residue class. For a nonzero-valuation dilation, finitely many affine comparisons select the eventual winner of each $R_h(q^n)$; project their constants modulo $\Delta_{|v(q)|}$ to obtain the coarsened bound (7.3), or (7.4) when Γ is divisible. When the parameter scale is noncofinal one can instead display $\gamma \gg |v(q)|$ and the explicit bad argument $t^{-(B+\gamma)}$ of Proposition 7.3.

These are mathematical certificate schemas relative to exact coefficient data and order comparisons. They are *not* a terminating algorithm for arbitrary encoded surreals. Deciding whether an arbitrary complex residue is a root of unity, comparing arbitrary surreal values, extracting arbitrary supports, or deciding whether an abstract scale is cofinal can require information not effectively available in a chosen representation, and none of this has been made computable by the proofs.

13.2 A valuation recurrence with zero coefficients allowed

Example 13.1 (Jump lengths, not just jumps). Suppose a sequence that is not eventually zero satisfies

$$t^A a_n + (n-2)t^B a_{n+1} + t^C a_{n+3} = 0 \quad (n \geq 0),$$

with $A, B, C \in \Gamma$. The middle coefficient vanishes at $n = 2$, a finite exceptional index; beyond it the three valuations are A, B, C . The escape chain may move forward by one or by three, and no assumption that all a_n are nonzero is needed. Corollary 3.4 excludes the region $v(x) \leq -\max\{A - B, A - C, 0\}$ under Convention 1.1 alone. If Γ is divisible, Corollary 3.5 gives the sharper excluded region

$$v(x) \leq -\max\left\{A - B, \frac{A - C}{3}\right\}.$$

The maximum must be divided by the actual jump length: replacing every jump by one need not preserve the intended threshold, especially when the valuation gaps have different signs. This is a worked instance of item (1) of Section 1.4.

13.3 Current Lean coverage and remaining dependency order

The ledger `docs/FORMALIZATION.md` records the formalized escape mechanism in `Surreal/HahnSeries/EscapeChain.lean`: the nonincreasing-order obstruction, escape step and infinite chain, evaluation exclusion, and the uniform bounded-cost corollary. It also covers the univariate strong-domain definitions and the formal exponential identity with its exact domain and attained exclusion boundary. Part (2) of Lemma 2.1 is mapped separately to `Surreal/HahnSeries/NeumannWords.lean`.

The next dependencies are integer valuation stabilization and the unit-orbit residue-polynomial argument, followed by the operator-to-recurrence conversions and the differential and dilation classifications. The refined periodic threshold, the complete support certificates, the new inward-stability lemma, the strengthened torsion equivalence and the generic-line extension remain unformalized. The several-variable proof also needs countable polynomial avoidance and restriction of a rational differential system.

Nothing in the nonlinear part, Sections 14–17, is formalized. The build and axiom audit cover only imported declarations recorded in the ledger. The Python checks are finite checks of formulas and examples; they do not establish any remaining infinite statement.

14 Nonlinear equations: finite initial polynomials

This section and the next three contain the material of source 10. They answer the first-order case of Question 19.1, which sources 08 and 09 had left open, together with a nondegenerate higher-order class and positive-weight Euler equations in several variables. The method is not the escape chain of Section 3. The warning in Remark 5.8 stands: a nonlinear equation gives coefficient convolutions, not a bounded-step linear recurrence. Instead, a whole scaled series is reduced to a finite polynomial at one large scale, and an exact leading-term computation gives the contradiction.

14.1 Coefficient field, conventions and the merge’s choices

Convention 14.1 (Coefficient field of the nonlinear part). In Sections 14–17, k is an arbitrary field of characteristic zero with the trivial valuation, Γ is as in Convention 1.1, and

$$\mathbb{K} = k((t^\Gamma)).$$

The valuation v , leading coefficient lc , residue res , strong summability, $\text{Dom}_{\text{str}}(f)$ and entireness are defined exactly as in Section 1.2 and Definition 1.2, with k in place of $\mathbb{C}$, except that, following source 10, $\text{res}(a)$ is defined for every a of nonnegative valuation as its coefficient at exponent zero, so that $\text{res}(a) = 0$ when $v(a) > 0$. Write $\mathcal{E}_\Gamma(k)$ for the strongly entire series in $\mathbb{K}[[z]]$. In several variables, strong entireness of $f = \sum_\alpha a_\alpha x^\alpha$ means strong summability of the *joint* family $(a_\alpha b^\alpha)_{\alpha \in \mathbb{N}^m}$ at every $b \in \mathbb{K}^m$. For $k = \mathbb{C}$ one has $\mathbb{K} = K_\Gamma$ and $\mathcal{E}_\Gamma(k) = \mathcal{E}_\Gamma$. The derivative D_z kills $\mathbb{K}$, and $f' = D_z f$; no scalar surreal derivation, analytic continuation or fine-topology limit is involved.

Why the coefficient field differs. Source 10 proves every statement of the nonlinear part over an arbitrary characteristic-zero field k , and it uses neither algebraic nor real closedness. Its hypothesis on the coefficient field is therefore weaker than the hypothesis $k = \mathbb{C}$ under which Theorems A–C and the rest of Sections 1–13 are stated. This generality is recorded for the nonlinear part only. It is *not* transferred to the linear theorems: they remain stated over $\mathbb{C}$, and their possible extension to other coefficient fields is the separate remark of Section 12, resting on its own argument. In the other direction, the nonlinear part borrows from Section 2 only facts about well-ordered subsets of Γ and finite coefficient sums (Lemmas 2.1 and 2.3), whose proofs do not use any property of $\mathbb{C}$; source 10 proves the facts it needs over k itself, except the positive-support lemma used in Theorem 15.10, which it imports as standard. The group Γ is not assumed divisible, and no order unit, no countable cofinal subset and no real-valued norm is used anywhere in the nonlinear part.

Renamed symbols. Several symbols of source 10 would collide with symbols of Sections 1–13; they are renamed here, in the new material only.

Source 10	Here	Reason
$D = \max_\nu d_\nu, W$	d_P, w_P	D_z is the derivative.
H_P (residue corner polynomial)	χ_P	H is an element of Γ in Sections 2–11.
B_P (degree cutoff, in $\mathbb{Q}$)	κ_P	B, B_L, B_{rec} are elements of Γ .
B (set of right sides)	$\mathcal{R}$	The same.
r, r_0 (scale, “radius”)	δ, δ_0	r indexes derivative orders in (1.2); $t^{-\delta}$ is the notation of Corollary 3.4.
h_r, p_r (Gauss value, initial polynomial)	$\gamma_\delta, \phi_\delta$	h is a torsion order, p_r an operator coefficient.
M (an integer bound)	n_*	$M(S)$ is a support monoid.
E_q (Euler operator)	ϑ_q	E_η is the formal exponential of Proposition 5.3.
Theorems A, B, C	Theorems D, 16.5, 16.1	Theorems A–C of this report are different theorems.

The shipped audit files `10-nonlinear-rigidity-PROOF_AUDIT.md` and `10-nonlinear-rigidity-SOURCES_AND_SCOPE.md` keep the source’s notation and numbering.

Words with two meanings. *Escape* keeps the meaning of Section 3 (a chain through a linear recurrence); source 10’s lemma “escape of active degree” is a different statement and is called Lemma 14.5 here. An *exclusion bound* is, as in Section 3, an element $\delta_0 \in \Gamma$ such that evaluation fails at the arguments $t^{-\delta}$ with $\delta \geq \delta_0$; the source’s word “radius” is not used. A *corner* is always the finite set of terms of Section 14.4, never a full differential Newton polygon. The *residue polynomials* $\overline{P}$ and R_r of Section 4 are unrelated to the residue corner polynomial χ_P .

Results printed once. Source 10 reproves several facts that Sections 2 and 11 already contain. They are printed once and cited: the support calculus inside Lemma 14.2 (Lemma 2.1 (1),(3) and Proposition 2.7); its “cofinal escape of coefficient minima” lemma, which is Lemma 2.3 (a); the positive-support geometric-series lemma, which source 10 imports and Lemma 2.1 (2) proves; the surreal embedding, which is Section 2.4 with the same monomial convention $t^\gamma \mapsto \omega^{-\gamma}$; strong entireness of the series (11.2), which is Theorem 11.2; and the remark on $\Gamma = 0$, which is in Section 12. Everything else in source 10 is printed in Sections 14–17, with its proofs.

14.2 The coefficient-family algebra

For a finite tuple $X = (X_1, \dots, X_m)$ of indeterminates put

$$\mathcal{S}_m = \left\{ F = \sum_{\alpha} b_{\alpha} X^{\alpha} \in \mathbb{K}[[X]] : (b_{\alpha})_{\alpha} \text{ is strongly summable} \right\},$$

and let $\mathcal{S}_m^+$ be the subset whose coefficients all have nonnegative valuation. The condition concerns the family of coefficients, not convergence in the X -adic topology.

Lemma 14.2 (Hahn-family algebra and its residue). *$\mathcal{S}_m$ is a $\mathbb{K}$ -subalgebra of $\mathbb{K}[[X]]$, stable under the formal partial derivatives and under the Euler operators $\sum_{\nu} q_{\nu} X_{\nu} \partial_{\nu}$ with $q_{\nu} \in k$. Coefficientwise residue is a ring homomorphism*

$$\rho : \mathcal{S}_m^+ \longrightarrow k[X], \quad \rho\left(\sum_{\alpha} b_{\alpha} X^{\alpha}\right) = \sum_{\alpha} \text{res}(b_{\alpha}) X^{\alpha},$$

whose values are polynomials, not arbitrary formal power series; ρ commutes with these differential operators.

Proof. Closure under sums, scalar multiples and products is the support calculation of Lemma 2.1 (1) and (3), as in the proof of Proposition 2.7. For strongly summable (b_{α}) and (c_{ϵ}) a fixed exponent has finitely many representations as a sum of two support exponents, each carried by finitely many indices, so the double family $(b_{\alpha} c_{\epsilon})$ is strongly summable, and regrouping by $\alpha + \epsilon$ gives the Cauchy product. Source 10 proves this over k by the same two-coordinate sequence argument; like the Hahn-field construction itself, that argument uses ordinary set-theoretic choice. A formal derivative or an Euler operator with coefficients in k reindexes a subfamily and multiplies its members by elements of the trivially valued field k ; it creates no new support exponent. For $F \in \mathcal{S}_m^+$ the exponent zero lies in only finitely many coefficient supports, so $\rho(F)$ is a polynomial. Each X -coefficient of a product is a finite sum of products of coefficients, the residue is additive and multiplicative on elements of nonnegative valuation, and the local finiteness just proved justifies every interchange; so ρ is a ring homomorphism. Differentiation commutes with residue coefficientwise. $\square$

For $0 \neq F = \sum_{\alpha} b_{\alpha} X^{\alpha} \in \mathcal{S}_m$ put

$$\gamma(F) = \min_{b_{\alpha} \neq 0} v(b_{\alpha}), \quad \text{in}(F) = \rho(t^{-\gamma(F)} F).$$

The minimum exists because it is the least element of the nonempty well-ordered union of supports. Then $t^{-\gamma(F)} F \in \mathcal{S}_m^+$ and the *initial polynomial* $\text{in}(F)$ is nonzero.

Corollary 14.3 (Gauss multiplicativity). *For nonzero $F, G \in \mathcal{S}_m$,*

$$\gamma(FG) = \gamma(F) + \gamma(G), \quad \text{in}(FG) = \text{in}(F) \text{in}(G).$$

Proof. Normalize both factors and apply Lemma 14.2. The product of the two nonzero residues is nonzero because $k[X]$ is an integral domain, so some coefficient of $t^{-\gamma(F)-\gamma(G)} FG$ has valuation exactly zero. $\square$

14.3 Scaled initial polynomials

Let $f = \sum_{n \geq 0} a_n z^n \in \mathbb{K}[[z]]$ and $\delta \in \Gamma$ with $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$. Then

$$F_\delta(X) = f(t^{-\delta} X) = \sum_{n \geq 0} a_n t^{-n\delta} X^n \in \mathcal{S}_1.$$

For $f \neq 0$ put

$$\gamma_\delta = \min_{a_n \neq 0} (v(a_n) - n\delta), \quad g_\delta = t^{-\gamma_\delta} F_\delta, \quad \phi_\delta = \rho(g_\delta), \quad N_\delta = \deg \phi_\delta. \quad (14.1)$$

The *active* indices are the n attaining γ_δ ; there are finitely many, N_δ is the largest, and $A_\delta = \text{lc}(\phi_\delta) \in k^\times$.

The element γ_δ is a *coefficient Gauss value*. It need not equal $v(f(t^{-\delta}))$: $\phi_\delta(1)$ can vanish although $\phi_\delta \neq 0$. The proofs below use identities in $\mathcal{S}_1$ and in $k[X]$ only, never an identification of these two quantities, and never replace the initial polynomial by its value at 1.

Lemma 14.4 (Exact derivative scaling). *For every $j \geq 0$,*

$$(D_z^j f)(t^{-\delta} X) = t^{\gamma_\delta + j\delta} g_\delta^{(j)}(X), \quad \rho(g_\delta^{(j)}) = \phi_\delta^{(j)}. \quad (14.2)$$

If $N_\delta \geq j$, then $\phi_\delta^{(j)}$ has degree $N_\delta - j$ and leading coefficient $A_\delta(N_\delta)_j$.

Proof. The scaling identity is a formal coefficient computation: $D_X^j F_\delta(X) = t^{-j\delta} (D_z^j f)(t^{-\delta} X)$. The residue identity is Lemma 14.2. The falling factorial is a nonzero element of k for $N_\delta \geq j$ because k has characteristic zero. $\square$

Lemma 14.5 (Large active degrees). *Let f be nonpolynomial, let $n_* \in \mathbb{N}$, and choose $m > n_*$ with $a_m \neq 0$. There is $\delta_1 \in \Gamma_{>0}$ such that, whenever $\delta \geq \delta_1$ and $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$, every active index exceeds n_* . Moreover $\gamma_\delta \leq v(a_m) - m\delta$.*

Proof. Choose $\delta_1 > 0$ with

$$(m - n)\delta_1 > v(a_m) - v(a_n) \quad (0 \leq n \leq n_*, a_n \neq 0). \quad (14.3)$$

There are finitely many inequalities, and they need no division in Γ : take δ_1 positive and larger than the absolute values of all right-hand sides, since each multiplier $m - n$ is a positive integer. They persist for $\delta \geq \delta_1$ and say that each such low-index term has scaled valuation $v(a_n) - n\delta$ strictly larger than that of the m th term. The last assertion is the definition of γ_δ . $\square$

This is not a limit $N_\delta \rightarrow \infty$ along a countable cofinal sequence, which need not exist; it concerns a final segment of $\Gamma_{>0}$ and only those scales at which strong evaluation is available. It is also unrelated to the escape chain of Lemma 3.1: no recurrence is used.

14.4 The finite corner of a differential polynomial

Let P be a nonzero differential polynomial of order at most s , written in combined-monomial form

$$P(z, Y_0, \dots, Y_s) = \sum_{\nu \in J} c_\nu z^{k_\nu} \prod_{j=0}^s Y_j^{m_{\nu j}}, \quad (14.4)$$

with J finite, every $c_\nu \in \mathbb{K}^\times$, and identical monomials already combined. Write $P[f] = P(z, f, D_z f, \dots, D_z^s f)$. Define the integers

$$d_\nu = \sum_{j=0}^s m_{\nu j}, \quad w_\nu = k_\nu - \sum_{j=0}^s j m_{\nu j}, \quad d_P = \max_\nu d_\nu, \quad w_P = \max_{d_\nu=d_P} w_\nu, \quad (14.5)$$

and

$$\mathcal{C}_P = \{\nu : d_\nu = d_P, w_\nu = w_P\}, \quad \beta_P = \min_{\nu \in \mathcal{C}_P} v(c_\nu), \quad \bar{c}_\nu = \text{res}(t^{-\beta_P} c_\nu) \in k \quad (\nu \in \mathcal{C}_P).$$

Thus (d_P, w_P) is chosen lexicographically: first the largest total differential degree, then the largest shift w . This is a finite corner selection, not the construction of a differential Newton polygon. The *residue corner polynomial* and the *full corner polynomial* are

$$\chi_P(T) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu \prod_{j=0}^s (T)_{\underline{j}}^{m_{\nu j}} \in k[T], \quad I_P(T) = \sum_{\nu \in \mathcal{C}_P} c_\nu \prod_{j=0}^s (T)_{\underline{j}}^{m_{\nu j}} \in \mathbb{K}[T]. \quad (14.6)$$

In general χ_P can vanish although some $\bar{c}_\nu \neq 0$, because different products of falling factorials can be linearly dependent. If $\chi_P \neq 0$, then $I_P \neq 0$, and

$$I_P(n) = 0 \implies \chi_P(n) = 0 \quad (n \in \mathbb{N}), \quad (14.7)$$

because $t^{-\beta_P} I_P$ has coefficients of nonnegative valuation and residue commutes with substitution of an ordinary integer.

Lemma 14.6 (The exact corner identity). *Let $\phi \in k[X]$ have degree $N \geq s$ and leading coefficient $A \neq 0$, and put*

$$\Omega_\phi(X) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{k_\nu} \prod_{j=0}^s (\phi^{(j)}(X))^{m_{\nu j}}.$$

Then $\deg \Omega_\phi \leq d_P N + w_P$ and

$$[X^{d_P N + w_P}] \Omega_\phi = A^{d_P} \chi_P(N). \quad (14.8)$$

In particular $\chi_P(N) \neq 0$ implies $\Omega_\phi \neq 0$.

Proof. Each corner summand has degree at most $k_\nu + \sum_j m_{\nu j}(N - j) = d_P N + w_P$, and its coefficient in that degree is $\bar{c}_\nu A^{d_P} \prod_j (N)_{\underline{j}}^{m_{\nu j}}$. Summing gives the formula. $\square$

Only finitely many nonnegative integers are roots of a nonzero χ_P , even when k is not algebraically closed. The active degree is an ordinary integer, not an element of Γ , and no root extraction or division in Γ occurs in this step.

15 First-order algebraic differential equations

15.1 A finite certificate excluding all large scales

The following theorem carries the proof of Theorem D and is also a practical audit object: its scale inequalities use only finitely many coefficients of the solution. It applies to equations of any order.

Theorem 15.1 (Nonlinear finite exclusion certificate). *Let P be as in (14.4) with $d_P \geq 1$ and $\chi_P \neq 0$, and let $f = \sum_n a_n z^n$ be a nonpolynomial formal solution of $P[f] = 0$. Choose an integer $n_* \geq s$ at least as large as every nonnegative integer root of χ_P , and choose $m > n_*$ with $a_m \neq 0$ and*

$$(d_P - d_\nu)m + w_P - w_\nu > 0 \quad (d_\nu < d_P). \quad (15.1)$$

Every $\delta \in \Gamma_{>0}$ satisfying the finitely many strict inequalities

$$(m - n)\delta > v(a_m) - v(a_n) \quad (0 \leq n \leq n_*, a_n \neq 0), \quad (15.2)$$

$$(w_P - w_\nu)\delta > \beta_P - v(c_\nu) \quad (d_\nu = d_P, w_\nu < w_P), \quad (15.3)$$

$$((d_P - d_\nu)m + w_P - w_\nu)\delta > \beta_P - v(c_\nu) + (d_P - d_\nu)v(a_m) \quad (d_\nu < d_P) \quad (15.4)$$

has $t^{-\delta} \notin \text{Dom}_{\text{str}}(f)$. Such m and δ exist, and the inequalities persist when δ increases. Consequently $\text{Dom}_{\text{str}}(f)$ omits $t^{-\delta}$ for every δ in a final segment of $\Gamma_{>0}$, and every strongly entire solution of $P[f] = 0$ is a polynomial.

Proof. An integer n_* exists because $\chi_P \neq 0$. For $d_\nu < d_P$ the coefficient $d_P - d_\nu$ of m in (15.1) is a positive integer, so all sufficiently large m satisfy it, and a nonpolynomial f has nonzero coefficients of arbitrarily large index. Every multiplier on the left of (15.2)–(15.4) is then a positive integer, so any δ that is positive and larger than the absolute values of all right-hand sides satisfies them, as does every larger δ .

Suppose $t^{-\delta} \in \text{Dom}_{\text{str}}(f)$ for such a δ , and form $\gamma = \gamma_\delta$, $g = g_\delta$, $\phi = \phi_\delta$, $N = N_\delta$ as in (14.1). By (15.2) and Lemma 14.5,

$$N > n_*, \quad \gamma \leq v(a_m) - m\delta. \quad (15.5)$$

All derivatives of g lie in $\mathcal{S}_1^+$. Substituting $z = t^{-\delta}X$, a ring homomorphism $\mathbb{K}[[z]] \rightarrow \mathbb{K}[[X]]$, Lemma 14.4 turns the ν th summand of $P[f]$ into

$$c_\nu t^{d_\nu \gamma - w_\nu \delta} X^{k_\nu} \prod_j (g^{(j)})^{m_{\nu j}}.$$

Multiply the whole identity by $t^{-\lambda}$ with $\lambda = \beta_P + d_P \gamma - w_P \delta$. The scalar in front of the ν th product then has valuation

$$v(c_\nu) - \beta_P + (d_\nu - d_P)\gamma + (w_P - w_\nu)\delta. \quad (15.6)$$

For $\nu \in \mathcal{C}_P$ this is $v(c_\nu) - \beta_P \geq 0$. For $d_\nu = d_P$ and $w_\nu < w_P$ it is positive by (15.3). For $d_\nu < d_P$, multiplying the bound on γ in (15.5) by the negative integer $d_\nu - d_P$ reverses it and gives the lower bound

$$v(c_\nu) - \beta_P - (d_P - d_\nu)v(a_m) + ((d_P - d_\nu)m + w_P - w_\nu)\delta > 0$$

by (15.4). So the normalized identity lies in $\mathcal{S}_1^+$, and only the corner summands survive the residue map. Applying ρ to the formal identity gives

$$0 = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{k_\nu} \prod_j (\phi^{(j)})^{m_{\nu j}} = \Omega_\phi(X). \quad (15.7)$$

But $N > n_* \geq s$ and $\chi_P(N) \neq 0$, so by Lemma 14.6 the coefficient of $X^{d_P N + w_P}$ in Ω_ϕ is $\text{lc}(\phi)^{d_P} \chi_P(N) \neq 0$, a contradiction. The last assertion follows because a strongly entire series is evaluable at every monomial argument. $\square$

An explicit, division-free exclusion bound. Let $\mathcal{R}$ be the finite set of right-hand sides of (15.2)–(15.4); an empty list contributes nothing. For any $\epsilon \in \Gamma_{>0}$,

$$\delta_0 = \epsilon + \max(\{0\} \cup \{|b| : b \in \mathcal{R}\}) \quad (15.8)$$

satisfies all of them, since their multipliers are positive integers. This is an explicit certificate; it is not claimed to be optimal. The proof uses the full supports to form the initial polynomial but only finitely many leading valuations to choose δ_0 . Leading valuations alone do *not* in general decide strong evaluation at a boundary argument, and the certificate has no converse.

For $k = \mathbb{C}$, Lemma 2.4 (inward stability) turns the monomial conclusion into the exterior form used in Theorem A: a nonpolynomial formal solution is then not strongly evaluable at any $x \neq 0$ with $v(x) \leq -\delta_0$. This combination of the two statements is made by the merge; source 10 states only the monomial form.

Corollary 15.2 (Cofinal scales suffice). *If $\chi_P \neq 0$, a formal solution of $P[f] = 0$ strongly evaluable at $t^{-\delta}$ for a cofinal set of $\delta \in \Gamma_{>0}$ is a polynomial.*

Proof. A cofinal subset meets the final segment excluded by Theorem 15.1 for a nonpolynomial solution. $\square$

Dependency audit. The only infinite-support step is Lemma 14.2, with the existence and finiteness of a leading coefficient layer; the rest is finite polynomial algebra. The proof makes none of the following replacements: a higher-rank group by a real-valued norm; strong summation by convergence of finite subsums; a nonsummable family by a cancelled expression; the residue polynomial by its value at 1; a nonlinear coefficient recurrence by a linear one. The three critical checkpoints are the minimum in (14.1), the order reversal when (15.5) is multiplied by $d_\nu - d_P < 0$, and the nonvanishing of $\chi_P(N)$.

15.2 Why the corner never degenerates in order one; proof of Theorem D

Write a first-order equation in combined-monomial form as

$$P(z, Y, Y') = \sum_{k,i,j} c_{kij} z^k Y^i (Y')^j. \quad (15.9)$$

For a term, $d = i + j$ and $w = k - j$. At the corner, fixing j forces $i = d_P - j$ and $k = w_P + j$, so there is at most one corner term for each j , and (14.6) becomes

$$\chi_P(T) = \sum_{\substack{i+j=d_P \\ k-j=w_P}} \bar{c}_{kij} T^j, \quad I_P(T) = \sum_{\substack{i+j=d_P \\ k-j=w_P}} c_{kij} T^j. \quad (15.10)$$

At least one $\bar{c}_{kij}$ is nonzero by the choice of β_P , and its power T^j cannot cancel against another summand. Hence $\chi_P \neq 0$.

Proof of Theorem D. If $d_P = 0$, then P is a nonzero polynomial in z alone and has no formal solutions. Otherwise the preceding observation and Theorem 15.1 give polynomiality and the exclusion statement, with δ_0 as in (15.8). A nonzero algebraic relation between f and $D_z f$ over $\mathbb{K}(z)$ becomes a nonzero polynomial of the form (15.9) after denominators are cleared, which gives the independence statement. $\square$

The uniqueness of a corner term for fixed j is a structural fact of order one. It is exactly what fails when several derivative orders can contribute the same falling-factorial polynomial (Section 16.2).

15.3 Finite candidate degrees and the entire-solution locus

Once polynomiality is known, ordinary polynomial degree gives a sharper finite candidate set that uses the *full* coefficients. Assume $d_P \geq 1$ and define the nonnegative rational number

$$\kappa_P = \max\left(\{0\} \cup \left\{\frac{w_\nu - w_P}{d_P - d_\nu} : d_\nu < d_P\right\}\right) \in \mathbb{Q}. \quad (15.11)$$

This divides ordinary integers in $\mathbb{Q}$, not elements of Γ .

Theorem 15.3 (Finite candidate degrees). *For the first-order polynomial (15.9) with $d_P \geq 1$, every nonconstant strongly entire solution has degree in the finite set*

$$\mathcal{D}_P = \{n \in \mathbb{Z}_{>0} : n \leq \kappa_P\} \cup \{n \in \mathbb{Z}_{>0} : I_P(n) = 0\}. \quad (15.12)$$

The constant solutions are exactly the $c \in \mathbb{K}$ with $P(z, c, 0) = 0$ as a polynomial in z . The set (15.12) is necessary; it does not promise that each candidate occurs.

Proof. Polynomiality is Theorem D. Let $f = az^n + \dots$ with $n \geq 1$ and $a \neq 0$. A term of (15.9) has degree $k + in + j(n-1) = dn + w$ and leading coefficient $c_{kij}a^d n^j$. If $n > \kappa_P$, every term with $d < d_P$ has degree below $d_P n + w_P$, and so do the terms with $d = d_P$, $w < w_P$. So the coefficient of $z^{d_P n + w_P}$ in $P[f]$ is $a^{d_P} I_P(n)$, which must vanish. I_P is nonzero by (15.10), so its set of ordinary integer roots is finite. The assertion on constants is immediate by substitution. $\square$

Remark 15.4 (What “finite” does and does not compute). The set of ordinary integer roots of I_P is well defined and finite for arbitrary Hahn coefficients, but its existence is not a uniform algorithm for arbitrary coefficient names: extracting a leading exponent, deciding coefficient equality, or testing an arbitrary complex coefficient for exact integrality can need information that no numerical approximation supplies. For rational coefficients, or other exact representations with the needed operations, (15.12) is an actual finite algebraic search. The shipped program uses only exact rational data.

Replacing $I_P(n) = 0$ by $\chi_P(n) = 0$ gives a coarser necessary set, by (14.7). Keeping I_P can be genuinely sharper.

Example 15.5 (A resonance visible only in the first layer). Let $c = t^\eta$ with $\eta > 0$, $N \in \mathbb{Z}_{>0}$, and $P = zY' - (N + c)Y$. Then $\chi_P(T) = T - N$ but $I_P(T) = T - N - c$ has no ordinary nonnegative integer root. The only formal solution in $\mathbb{K}[[z]]$ is zero, since coefficient comparison gives $(n - N - c)a_n = 0$ for every n . A residue resonance alone is not a polynomial solution.

Corollary 15.6 (An algebraic entire-solution locus). *For a first-order equation P with $d_P \geq 1$ put $n_P = \max(\{0\} \cup \mathcal{D}_P)$. There is an explicitly defined affine algebraic set $\mathcal{V}_P \subseteq \mathbb{A}_{\mathbb{K}}^{n_P+1}$ whose $\mathbb{K}$ -points are in bijection with all strongly entire formal solutions of $P[f] = 0$. The same equations parametrize all strongly entire solutions over every Hahn-field extension $\mathbb{K} \hookrightarrow \mathbb{L} = k'((t^{\Gamma'}))$ induced by an embedding of coefficient fields $k \hookrightarrow k'$ and an embedding of ordered groups $\Gamma \hookrightarrow \Gamma'$.*

Proof. With indeterminates $u_0, \dots, u_{n_P}$ put $F_u(z) = \sum_{n \leq n_P} u_n z^n$ and expand $P(z, F_u, F'_u) = \sum_\ell Q_\ell(u) z^\ell$ with $Q_\ell \in \mathbb{K}[u_0, \dots, u_{n_P}]$. Let $\mathcal{V}_P$ be the common zero locus of these finitely many Q_ℓ . Its points are exactly the polynomial solutions of degree at most n_P , each of which is strongly entire, and Theorem 15.3 shows there are no others. Under the stated extension the monomial degrees and κ_P are unchanged, and for an ordinary integer n the element $I_P(n)$ vanishes in $\mathbb{L}$ if and only if it vanishes in $\mathbb{K}$. Theorem D applies over $\mathbb{L}$, whose coefficient field again has characteristic zero and whose group is again nonzero, so the same degree bound and equations remain exhaustive there. $\square$

This is a finite-dimensional algebraic parametrization, not a claim that the number of solutions is finite, and not a moduli statement over coefficient rings with nilpotents. The extension class is exactly the one stated.

15.4 Worked first-order equations

All identities use the formal derivative D_z . For $k = \mathbb{C}$ they are statements about surcomplex Hahn workspaces; for $k = \mathbb{R}$ the real versions follow with the non-real constant roots omitted. Consider

$$D_z f = f^2 + p(z), \quad p \in \mathbb{K}[z]. \quad (15.13)$$

The corner is the term $-Y^2$, so $d_P = 2$, $w_P = 0$ and $I_P = -1$. If p has degree $e \geq 0$, then $\kappa_P = \max(0, e/2)$, so a nonconstant entire solution has degree at most $\lfloor e/2 \rfloor$. More precisely, a solution of degree $n \geq 1$ requires $\deg p = 2n$ and the leading coefficient of p to be minus the square of that of f , since the top term of f^2 cannot be cancelled by $D_z f$. These conditions are necessary only; the remaining coefficient equations must still hold.

Proposition 15.7 (Two exact Riccati classifications). *Over $\mathbb{K} = \mathbb{C}((t^\Gamma))$, the strongly entire solutions of $D_z f = f^2 + 1$ are exactly $f = i$ and $f = -i$, and the only strongly entire solution of $D_z f = f^2 + 1 - z^2$ is $f = z$.*

Proof. In the first equation the degree bound excludes nonconstant solutions, and a constant must satisfy $c^2 + 1 = 0$; in the field $\mathbb{K}$, which contains i , the roots are $\pm i$. In the second the degree bound is one and constants are impossible. For $f = az + b$ coefficient comparison gives $a^2 = 1$, $2ab = 0$ and $a = b^2 + 1$, so $b = 0$ and $a = 1$, and $f = z$ is verified directly. $\square$

The first equation is also `diff:cor:riccati` of the collection's differential-equations report [13], with the same two solutions, but there $\partial u = 1 + u^2$ is solved in $\mathbf{No}[i]$ for a derivation on surreal scalars. The two statements concern different derivations and different objects, and neither implies the other.

Example 15.8 (The degree cutoff is attained). For every integer $m \geq 1$ the equation $D_z f = f^2 + mz^{m-1} - z^{2m}$ has the strongly entire solution $f = z^m$. Here $I_P = -1$ and $\kappa_P = m$ exactly, so the low-degree part of (15.12) cannot in general be replaced by $n < \kappa_P$.

Example 15.9 (Arbitrarily large resonant degrees). For $N \geq 1$ the equation

$$zf D_z f - Nf^2 = 0 \quad (15.14)$$

has $d_P = 2$, $w_P = 0$ and $I_P(T) = T - N$. Since $\mathbb{K}[[z]]$ is an integral domain, a nonzero solution satisfies $zD_z f = Nf$, and coefficient comparison gives exactly $f = cz^N$ with $c \in \mathbb{K}$ (including $c = 0$). The exceptional integer root is an actual degree, and no degree bound can depend only on the positions of the monomials of P : the coefficient N matters.

15.5 The exclusion bound cannot be uniform

Theorem A has an exclusion bound depending only on the operator. For nonlinear equations this is false, already for one fixed equation of first order.

Theorem 15.10 (A fixed Riccati equation defeats every uniform bound). *For $\lambda \in \Gamma$ let $c = t^\lambda$ and*

$$f_\lambda(z) = \frac{c}{1 - cz} = \sum_{n \geq 0} c^{n+1} z^n \in \mathbb{K}[[z]].$$

Then $D_z f_\lambda = f_\lambda^2$, f_λ is nonpolynomial, and

$$\text{Dom}_{\text{str}}(f_\lambda) = \{0\} \cup \{x \in \mathbb{K}^\times : v(x) + \lambda > 0\}. \quad (15.15)$$

Consequently, for every $\delta \in \Gamma$ some nonpolynomial solution of the fixed equation $D_z f = f^2$ is strongly evaluable at $t^{-\delta}$, and no exclusion bound depending on the equation alone excludes all its nonpolynomial solutions.

Proof. The derivative identity is a coefficient comparison. For $x \neq 0$ put $u = cx$, so the n th evaluation term is cu^n . If $v(u) > 0$, write $u = bt^{v(u)}(1+w)$ with $b \in k^\times$ and w zero or of positive valuation; then $\text{supp}(u^n) \subseteq nv(u) + M(\text{supp } w)$, the leading exponents $nv(u)$ increase strictly, and Lemma 2.3 (b), resting on the positive-support monoid of Lemma 2.1 (2), gives strong summability; multiplication by c preserves it. (Source 10 cites the Hahn–Neumann positive-support lemma here instead of proving it; no Archimedean convergence of $nv(u)$ is used.) If $v(u) = 0$, every cu^n has the nonzero coefficient $\text{lc}(u)^n$ at exponent λ , so local finiteness fails. If $v(u) < 0$, the leading exponents $\lambda + nv(u)$ strictly decrease. At $x = 0$ only one member is nonzero. This proves (15.15). Given δ , choose $\lambda > \delta$; then $v(t^{-\delta}) + \lambda = \lambda - \delta > 0$. $\square$

There is no conflict with Theorem 15.1, whose finite data, and hence whose bound, may depend on the solution; for f_λ the exact boundary moves with λ . At the boundary $v(x) = -\lambda$ geometric-series algebra cannot legalize infinitely many contributions at one exponent. Nor is there a conflict with Theorem A: each f_λ also satisfies the linear equation $(1 - t^\lambda z)D_z f - t^\lambda f = 0$, whose operator depends on λ , exactly as the operator $D_z - t^\eta$ of the formal exponential of Proposition 5.3 depends on η . (This comparison is the merge’s; it is a direct computation.)

16 Higher order and positive-weight Euler equations

16.1 The nondegenerate higher-order class

Theorem 16.1 (Higher-order corner criterion and its boundary). *Let P be a nonzero differential polynomial (14.4) of any finite order with $d_P \geq 1$ and $\chi_P \neq 0$. Then every strongly entire formal solution of $P[f] = 0$ is a polynomial, and every nonpolynomial formal solution fails strong evaluation at $t^{-\delta}$ for all δ in a final segment of $\Gamma_{>0}$, with the finite certificate of Theorem 15.1. The condition $\chi_P \neq 0$ holds automatically in order one, but not in order two: a fixed second-order equation can have polynomial solutions of every degree (Proposition 16.4).*

Proof. The first two assertions are Theorem 15.1; the third is Section 15.2 and Proposition 16.4. $\square$

Proposition 16.2 (Higher-order degree candidates). *Suppose $d_P \geq 1$ and $\chi_P \neq 0$ for an equation of order at most s . Every nonzero strongly entire solution has degree in the finite set*

$$\{0, \dots, s-1\} \cup \{n \in \mathbb{N} : n \geq s, n \leq \kappa_P\} \cup \{n \in \mathbb{N} : n \geq s, I_P(n) = 0\},$$

with κ_P as in (15.11). When $s = 0$ the first set is empty. Constant solutions can always be checked by substitution.

Proof. Polynomiality is Theorem 15.1. For degree $n \geq s$ every derivative leading-term formula of Lemma 14.6 applies, and if also $n > \kappa_P$ the top coefficient of $P[f]$ is $\text{lc}(f)^{d_P} I_P(n)$, which must vanish; $I_P \neq 0$ because $\chi_P \neq 0$. Degrees below s , where derivative leading terms can vanish, are listed separately. $\square$

Proposition 16.3 (The first two Painlevé polynomial equations). *In characteristic zero, the equation $D_z^2 f = 6f^2 + z$ has no strongly entire formal solution, and the equation $D_z^2 f = 2f^3 + zf + \alpha$ with $\alpha \in \mathbb{K}$ has a strongly entire solution exactly when $\alpha = 0$, the only one then being $f = 0$.*

Proof. The first equation has $d_P = 2$, $w_P = 0$, $\chi_P = -6$; the second has $d_P = 3$, $w_P = 0$, $\chi_P = -2$. Theorem 15.1 makes every entire solution a polynomial. In the first equation a polynomial of degree $n \geq 1$ gives f^2 of degree $2n$, larger than $\deg D_z^2 f$ (when nonzero) and than $\deg z = 1$, so no cancellation is possible, and a constant cannot cancel z . In the second, for $n \geq 1$ one has $3n > n + 1$ and $3n > \deg D_z^2 f$, so $2f^3$ cannot cancel; a constant c must satisfy $2c^3 + zc + \alpha = 0$ identically, which forces $c = 0$ and $\alpha = 0$, and substitution proves sufficiency. $\square$

The names only identify the displayed equations; no classification of their classical meromorphic solutions is imported. The examples illustrate a class beyond first order: a unique term of greatest total differential degree has a nonzero corner whenever its falling-factorial product is nonzero. In particular an equation $D_z^s f = Q(z, f)$ with Q of degree at least two in f has a nonzero corner and hence only polynomial strongly entire solutions; the coefficient of the top power of f may be a nonconstant polynomial in z , whose highest z -term then selects the corner.

16.2 A vanishing corner

Consider

$$z^2 f D_z^2 f + z f D_z f - z^2 (D_z f)^2 = 0. \quad (16.1)$$

All three terms have $d = 2$ and $w = 0$, and the residue corner polynomial is

$$T(T - 1) + T - T^2 = 0, \quad (16.2)$$

an identity in $k[T]$, not an exceptional integer resonance.

Proposition 16.4 (Complete formal classification of the example). *The formal solutions of (16.1) in $\mathbb{K}[[z]]$ are exactly 0 and cz^n with $c \in \mathbb{K}^\times$ and $n \in \mathbb{N}$. All are strongly entire, and their degrees are unbounded for this single equation.*

Proof. With $\vartheta = zD_z$, equation (16.1) reads $f \vartheta^2 f - (\vartheta f)^2 = 0$. For $f \neq 0$ work in the Laurent series field $\mathbb{K}((z))$ and divide by f^2 : $\vartheta(\vartheta f/f) = 0$. The constants of ϑ in $\mathbb{K}((z))$ are exactly $\mathbb{K}$, since $\vartheta \sum b_n z^n = \sum n b_n z^n$ and characteristic zero kills no nonzero integer. Hence $\vartheta f/f = c_0 \in \mathbb{K}$; if the first nonzero term of f has exponent n , comparing lowest terms gives $c_0 = n$. Then $\vartheta f = n f$ kills every coefficient except the n th. Substitution verifies the monomials and zero. $\square$

The example has two messages. A universal higher-order degree bound cannot be obtained by declaring χ_P nonzero: it may vanish identically, and a fixed equation can then have unbounded degrees. And the example is *not* a nonpolynomial entire solution, so it does not refute polynomial rigidity in higher order; it locates a failure of this proof and of a stronger uniform conclusion.

The corner is also not an invariant of the solution set. Multiplying an equation by another differential polynomial, or passing to a differential consequence, can change the corner data. The

criterion $\chi_P \neq 0$ is a sufficient condition on the displayed combined polynomial: when $\chi_P = 0$ another equation satisfied by the same function may have a nonzero corner, or a logarithmic-derivative reduction as in Proposition 16.4 may be available. A zero corner cannot be repaired by retaining only monomial degrees, since the residue coefficients and their exact cancellation are essential. The methods here do not decide whether a higher-order equation with vanishing residue corner has nonpolynomial strongly entire solutions; that is Question 19.1.

16.3 Positive-weight Euler equations in several variables

Fix positive integers $q = (q_1, \dots, q_m)$, write $\text{wt}(\alpha) = q \cdot \alpha$ for $\alpha \in \mathbb{N}^m$, and let

$$\vartheta_q = \sum_{\nu=1}^m q_\nu x_\nu \frac{\partial}{\partial x_\nu}, \quad \vartheta_q x^\alpha = \text{wt}(\alpha) x^\alpha.$$

Only finitely many multi-indices have weight at most a given integer; this, with the eigenvalue identity, is the essential feature of positive weights. Write a nonzero equation in combined-monomial form

$$P(x, Y, Z) = \sum_{\nu \in J} c_\nu x^{\alpha_\nu} Y^{i_\nu} Z^{j_\nu}, \quad (16.3)$$

to be read as $P(x, f, \vartheta_q f) = 0$, and set $d_\nu = i_\nu + j_\nu$, $w_\nu = \text{wt}(\alpha_\nu)$, with d_P , w_P , $\mathcal{C}_P$, β_P and $\bar{c}_\nu$ as in Section 14.4. The corner polynomials now keep the spatial variables:

$$\chi(T, X) = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{\alpha_\nu} T^{j_\nu}, \quad I(T, X) = \sum_{\nu \in \mathcal{C}_P} c_\nu X^{\alpha_\nu} T^{j_\nu}. \quad (16.4)$$

Both are nonzero: for fixed (α_ν, j_ν) the condition $d_\nu = d_P$ forces $i_\nu = d_P - j_\nu$, so no two combined corner terms give the same monomial in (X, T) . Only finitely many integers n make $\chi(n, X)$ the zero polynomial, since they must all be roots of one nonzero X -coefficient of χ , and likewise for I .

Theorem 16.5 (Positive-weight Euler rigidity). *Let $f \in \mathbb{K}[[x_1, \dots, x_m]]$ satisfy $P(x, f, \vartheta_q f) = 0$ for a nonzero $P \in \mathbb{K}[x_1, \dots, x_m, Y, Z]$. If f is nonpolynomial, it fails joint strong evaluation at $(t^{-q_1\delta}, \dots, t^{-q_m\delta})$ for every sufficiently large $\delta \in \Gamma_{>0}$. Consequently, if f is strongly entire, or merely jointly strongly evaluable at these points for a cofinal set of $\delta \in \Gamma_{>0}$, then f is a polynomial. Every fixed equation has a finite set of possible positive weighted degrees (Corollary 16.6).*

Proof. If $d_P = 0$, then P is a nonzero polynomial in x alone and there is no solution; assume $d_P \geq 1$. Choose $n_* \geq 0$ with $\chi(n, X) \neq 0$ for every integer $n > n_*$. Since $f = \sum_\alpha a_\alpha x^\alpha$ is nonpolynomial and bounded-weight sets are finite, choose $a_\mu \neq 0$ of weight $n_\mu = \text{wt}(\mu) > n_*$ so large that $(d_P - d_\nu)n_\mu + w_P - w_\nu > 0$ for $d_\nu < d_P$. Take $\delta > 0$ satisfying the finitely many inequalities

$$\begin{aligned} (n_\mu - \text{wt}(\alpha))\delta &> v(a_\mu) - v(a_\alpha) & (\text{wt}(\alpha) \leq n_*, a_\alpha \neq 0), \\ (w_P - w_\nu)\delta &> \beta_P - v(c_\nu) & (d_\nu = d_P, w_\nu < w_P), \\ ((d_P - d_\nu)n_\mu + w_P - w_\nu)\delta &> \beta_P - v(c_\nu) + (d_P - d_\nu)v(a_\mu) & (d_\nu < d_P); \end{aligned}$$

as before they can be met in every nonzero ordered group and persist on a final segment. Suppose the joint family is strongly summable at such a scale, and put

$$F(X) = f(t^{-q_1\delta} X_1, \dots, t^{-q_m\delta} X_m) \in \mathcal{S}_m, \quad \gamma = \gamma(F), \quad g = t^{-\gamma} F, \quad \phi = \rho(g).$$

Then $\gamma \leq v(a_\mu) - n_\mu \delta$, and the first inequalities show that ϕ has no monomial of weight at most n_* ; its greatest weight N exceeds n_* . Let $\phi_N \neq 0$ be its weight- N homogeneous part. The Euler operator commutes with the scaling, so $(\vartheta_q f)(t^{-q_1 \delta} X_1, \dots, t^{-q_m \delta} X_m) = t^\gamma \vartheta_q g$. Normalizing the substituted equation by $t^{-\beta_P - d_P \gamma + w_P \delta}$, the scalar computation (15.6) shows that only corner terms survive the residue, whence

$$0 = \sum_{\nu \in \mathcal{C}_P} \bar{c}_\nu X^{\alpha_\nu} \phi^{i_\nu} (\vartheta_q \phi)^{j_\nu}. \quad (16.5)$$

The largest possible weight on the right is $d_P N + w_P$. Since the weight- N part of $\vartheta_q \phi$ is $N \phi_N$, the homogeneous part of that weight is

$$\phi_N(X)^{d_P} \chi(N, X), \quad (16.6)$$

a product of two nonzero polynomials, hence nonzero. This contradicts (16.5). $\square$

The proof uses *joint* strong evaluation, not a scalar expression obtained by first setting $x_1 = \dots = x_m$ and cancelling: a factor $x_1 - x_2$ can make a diagonal scalar expression zero while the ungrouped monomial family is not summable. Keeping $X_1, \dots, X_m$ in (16.6) avoids that error.

Corollary 16.6 (Weighted-degree candidates). *With κ_P defined by (15.11) from the Euler data d_ν, w_ν , every nonconstant strongly entire solution of $P(x, f, \vartheta_q f) = 0$ has greatest weight in the finite set*

$$\{n \in \mathbb{Z}_{>0} : n \leq \kappa_P\} \cup \{n \in \mathbb{Z}_{>0} : I(n, X) \equiv 0\}.$$

For a fixed equation all strongly entire solutions are therefore described by finitely many polynomial equations in the coefficients of a bounded-weight polynomial ansatz.

Proof. Polynomiality is Theorem 16.5. For a polynomial of greatest weight $N > \kappa_P$, the terms of lower differential degree have smaller total weight, and the top-weight part of the equation is $f_N^{d_P} I(N, X)$, which must vanish; so $I(N, X) = 0$. The finite-root argument after (16.4) applies to I , and a bounded-weight ansatz has finitely many coefficients. $\square$

Example 16.7 (An exactly described solution space). For $N \geq 0$ the solutions of $\vartheta_q f = Nf$ in $\mathbb{K}[[x]]$ are exactly the weighted homogeneous polynomials of weight N , since coefficient comparison gives $(\text{wt}(\alpha) - N)a_\alpha = 0$. For $q = (1, 2)$ this space has dimension $\lfloor N/2 \rfloor + 1$ over $\mathbb{K}$, with basis $x_1^{N-2j} x_2^j$, $0 \leq j \leq \lfloor N/2 \rfloor$. The nonlinear equation $f(\vartheta_q f - Nf) = 0$ has the same solutions, because $\mathbb{K}[[x]]$ is an integral domain.

Why positive weights are needed. Let $\mathbb{K} = \mathbb{C}((t^{\mathbb{Q}}))$ and $G(u) = \sum_{n \geq 0} t^{n^2} u^n$, which is strongly entire (Corollary 17.1). The function $f(x, y) = G(y)$ is nonpolynomial and strongly entire but satisfies $x \partial_x f = 0$, so a zero weight invalidates the theorem. For the mixed weights $(1, -1)$, the nonpolynomial strongly entire function $f(x, y) = G(xy)$ satisfies $(x \partial_x - y \partial_y) f = 0$; at a fixed point its nonzero joint evaluation terms are exactly $t^{n^2} (xy)^n$, which are strongly summable. Positivity is therefore not a cosmetic substitute for nonzero weights, and a general first-order partial differential equation need not have the rigidity of a positive Euler operator.

Relation to Theorem C. Theorem C assumes that *all* mixed partial derivatives span a finite-dimensional space, and Section 6 notes that a single differential equation in several variables cannot force polynomiality in general, since it may ignore variables; the zero-weight example above is an instance. Theorem 16.5 instead assumes one algebraic relation between f and $\vartheta_q f$ with all weights positive, and no finite-dimensional span. Neither theorem implies the other.

17 Nonlinear consequences, hypotheses and predecessors

17.1 Two-jet independence of explicit series

With the surreal embedding of Section 2.4, for instance $\mathbb{C}((\omega^{-\mathbb{Q}}))$ is a set-sized surcomplex workspace to which Theorem D applies, and its coefficients and admissible arguments are actual elements of $\mathbf{No}[i]$. The independence conclusions, however, concern *formal functions*: each value $f(x)$ lies in $\mathbb{K}$ and is of course not transcendental over $\mathbb{K}$.

Corollary 17.1 (Two independent jets of the quadratic-exponent series). *Over $\mathbb{K} = \mathbb{C}((t^{\mathbb{Q}}))$ let $G(z) = \sum_{n \geq 0} t^{n^2} z^n$. Then G is nonpolynomial and strongly entire, and G and $D_z G$ are algebraically independent over $\mathbb{K}(z)$.*

Proof. For $x \neq 0$ with $v(x) = \sigma \in \mathbb{Q}$ the n th evaluated member has valuation $n^2 + n\sigma$, which eventually exceeds every rational bound, so Lemma 2.3 (a) gives strong summability even when x has an arbitrary infinite Hahn tail; at $x = 0$ the family is finite. All coefficients are nonzero. Theorem D gives the independence. $\square$

The series is not new: it is the rank-one example of the rank-one-Berkovich report [14], Example `found:ex:internal` of the foundations report [15], and a rescaled partial theta series: $G(z) = \Theta_{t^2}(tz)$ in the notation of (8.1). The contribution of source 10 is the nonlinear two-jet consequence only.

Corollary 17.2 (An example without an order unit). *In K_{Γ_∞} , with Γ_∞ as in (11.1), the strongly entire series $F = \sum_{n \geq 0} t^{e_n} z^n = \sum_{n \geq 0} \omega^{-\omega^n} z^n$ of (11.2) satisfies: F and $D_z F$ are algebraically independent over $K_{\Gamma_\infty}(z)$.*

Proof. Strong entireness is Theorem 11.2; source 10 proves it again by the cofinal-valuation route recorded in the second paragraph of that proof. Apply Theorem D. $\square$

F and its linear rigidity are in Theorem 11.2; the new statement is the two-jet strengthening. It answers the first-order case of the last sentence of Question 19.1 with no coefficient restriction, and says nothing about relations of order two or more.

Extension of scalars versus preservation of entireness. The degree classification is stable under the Hahn-field extensions of Corollary 15.6, but entireness of a nonpolynomial series need not be. Enlarge $\mathbb{Q}$ to $\mathbb{Q} + \mathbb{Q}\omega$ with its surreal order and view G over the larger field. At $z = t^{-\omega}$ the member exponents are $n^2 - n\omega$, with successive differences $(2n + 1) - \omega < 0$, so the support union contains an infinite descending sequence and the evaluation is not strong. The same computation appears in the rank-one-Berkovich report [14], and for F in Section 11.2. No theorem here claims that G or F is entire on the proper class $\mathbf{No}[i]$; hypotheses and derivative are relative to a specified set-sized workspace.

17.2 Which hypotheses are used

Example 17.3 (Positive characteristic destroys the conclusion). Let p be prime and $\mathbb{K} = \mathbb{F}_p((t^{\mathbb{Q}}))$. The nonpolynomial series $U(z) = \sum_{n \geq 0} t^{n^2} z^{pn}$ is strongly entire: at a point of valuation σ its member valuations are $n^2 + pn\sigma$, and Lemma 2.3 (a) applies, its proof being independent of the coefficient field. Yet $D_z U = 0$.

So the conclusion fails in positive characteristic even for one linear equation. The characteristic-zero requirements of the proof are substantive: ordinary integer eigenvalues stay distinct, nonzero falling factorials stay nonzero, and a nonzero polynomial cannot vanish at all ordinary integers.

The hypothesis $\Gamma \neq 0$ is needed for the scale argument; the degenerate case $\Gamma = 0$ is treated in Section 12. Under ordinary complex analytic summation, by contrast, $\exp z$ is a nonpolynomial entire function with $f' = f$; these are different notions of infinite sum, and the classical exponential does not contradict the Hahn theorem. The coefficient field may be $\mathbb{R}$, $\mathbb{C}$ or any field of characteristic zero; neither algebraic nor real closedness is used, no completeness of a real-valued norm is used, and Γ needs neither an order unit nor a countable cofinal subset. Strong summability supplies exactly the finite initial layer the argument needs.

17.3 An exact finite workflow

For first-order input P the reduction is as follows. Combine like monomials and remove zero coefficients. If no Y or Y' remains, decide the polynomial identity directly. Otherwise compute d_P , w_P , I_P and κ_P from the finite term list, find the positive integer roots of I_P in the chosen exact representation of the coefficients, take the largest candidate degree, substitute a generic polynomial of that degree, and solve the resulting finite algebraic system. This procedure is complete for *entire* solutions whenever its finite coefficient operations and algebraic solving are available. It is not a general algorithm for arbitrary surreal names, it does not enumerate the local formal solutions, and it does not decide arbitrary strong-support membership. A negative answer after an exhaustive exact solve is a genuine nonexistence certificate for entire solutions; failure of a numerical solver is not.

For a nonpolynomial candidate formal solution a second workflow builds the certificate (15.2)–(15.4) from one sufficiently high nonzero coefficient and finitely many low-index ones; a claimed certificate must supply their exact leading valuations. The shipped program checks finite algebraic identities and sample certificates; it does not decide infinitary summability.

17.4 Predecessors, repository comparison and priority

Inherited ideas. Hahn fields, the positive-support summation lemma, Conway normal forms, Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning are established ideas and are not relabelled as new. The organizing method is related in spirit to maximum-term and central-index arguments in the Wiman–Valiron tradition, surveyed by Hayman [18]; here the finite residue polynomial gives an exact identity rather than an analytic approximation, and no real-valued growth estimate is imported. Source 10 consulted Hayman’s survey through its publisher metadata and abstract only, and uses no theorem from its full text.

Newton-polygon methods for nonlinear equations have a substantial history. Cano and Fortuny Ayuso construct initial and indicial polynomials for nonlinear q -difference equations and obtain support and growth results for generalized-series solutions [19]. Their operator and setting are not those of a formal z -series with arbitrary-rank Hahn *coefficients* evaluated strongly at every workspace scale. The conceptual precedent is important, and no invention of differential Newton polygons is claimed.

Non-Archimedean value distribution already restricts nonlinear functional and differential equations. Hu and Luan work over a complete algebraically closed field with a nontrivial non-Archimedean absolute value and prove Clunie-, Malmquist- and Mokhon’ko-type results [20]; source 10 consulted

the parsed text of that paper for its field hypotheses and Theorems 1.1, 1.3 and 1.5, reports that a remote screenshot service failed so that no visual inspection of the PDF is claimed, and reproduces no figure or table from it. Hu and Yang’s monograph [21] is a broader predecessor; only its publisher metadata and description were checked, so no claim is made that all rank-one special cases were previously unknown. Giansiracusa and Mereta develop differential enhancements of non-Archimedean norms and a tropicalization of differential equations [22]; the finite corner here records only the data this obstruction needs, and no equivalence to their framework and no new tropical fundamental theorem is asserted. Mantova and Matusinski’s survey [17] was consulted for normal forms.

Repository comparison at source 10’s pin. Source 10 compared its results with the collection at 048b72c; the comparison, with the corrections needed now, is as follows.

Material	Existing result or scope	Difference in the nonlinear part
This report, Theorem A	Strongly entire D-finite series are polynomials; the nonlinear case was the open Question 15.1, now Question 19.1.	All first-order algebraic equations, a higher-order corner criterion, and solution-dependent finite certificates.
This report, Theorem C	Finite-dimensional span of all mixed derivatives forces polynomiality.	One algebraic relation in f and $\vartheta_q f$ suffices for positive weights; no finite-dimensional span.
Tail-spans report [16]	Particular Galois-supported constructions have all-order differential independence, for scalar or fixed-domain analytic derivations over specified smaller base fields.	A two-jet theorem for every nonpolynomial strongly entire function over its own full Hahn field.
Documentation catalogue docs/README.md	Distinguishes workspaces, strong summability, coefficient rings and scalar derivations.	Those distinctions are kept; no global proper-class analytic claim is added.

The tail-span examples are neither weakened nor rediscovered: that report proves stronger all-order independence for particular constructions under different coefficient and function-class hypotheses, while the present universal statement assumes strong entireness and concludes only two-jet independence. Source 10 cited the quadratic-exponent series as occurring in this report and in the catalogue; within the collection it occurs as the rank-one-Berkovich example [14], as `found:ex:internal` [15], and as a case of (8.1). Since that pin the collection has gained this nonlinear part itself; no other report now contains a nonlinear rigidity theorem for D_z , and the Riccati statement `diff:cor:riccati` of the differential-equations report concerns a scalar derivation (compare Proposition 15.7).

Source 10’s repository audit covered the root README, the catalogue, contents listings, this report’s definitions, theorems and nonlinear-question passages, and the README files of the tail-span, analytic-geometry and product-birthday reports (the last two for context only). It was a targeted comparison, not a line-by-line search of every report, Lean file, archive, branch or the Git history, and no absence claim rests on a truncated preview.

Priority and proof status. Source 10’s literature search, made on 22 September 2026, did not locate the exact all-rank strong-family first-order theorem, the finite exclusion certificate, or the positive-weight Euler formulation. Some searches were poorly targeted, and no exhaustive MathSciNet, zbMATH, full-book or citation-network review was made, so priority is provisional.

The safest description is a proposed new theorem package and a proved first-order answer to a repository question, which is not a named historical conjecture; it is not a certified first discovery of nonlinear non-Archimedean rigidity. The written proofs are the evidence. The finite checks of Section 18 establish no arbitrary-support theorem; no Lean formalization of the nonlinear part exists, and no independent human referee has checked it.

18 The three finite check suites

Each source manuscript shipped an independent exact-arithmetic suite. The suites of sources 08 and 09 were run on copies and passed, and neither supersedes the other: they cover overlapping but distinct finite facts of the linear theory, and they are kept side by side in `code/` for that reason. The suite of source 10 checks finite facts of the nonlinear part only; it was also run on a copy for this merge and reproduced its recorded counts exactly. All three use only the Python standard library, integers, and `fractions.Fraction`, with no floating point, no network access, and no third-party packages.

`code/08-finite-recurrences-verify.py` — 3,705 checks, 0 failures. It emphasises the operator-to-recurrence conversions in all three formats and the higher-rank ordered-group arithmetic.

Finite calculation	Checks
Differential operator to coefficient recurrence	648
Dilation operator to coefficient recurrence	336
Mixed operator to coefficient recurrence	276
Unit residues, parity, and finite exceptional cancellation	400
Partial theta identities and finite quadratic estimates	552
Higher-rank lexicographic valuation examples	1,292
Sparse Airy recurrence and finite escape edges	201
Total	3,705

The operator tests compare the grouped recurrence formulas against direct finite polynomial differentiation, dilation and multiplication; the theta tests compare Laurent monomial exponents exactly, including the q^{-1} -to- q conversion of (8.7); the unit tests retain the exceptional index $n = 7$ rather than silently checking only generic values; and the ordered-group checks use lexicographic integer tuples, not a real-valued surrogate norm. This script takes a *required* `--output` path and exits nonzero without one, which is deliberate: it prevents a re-run from overwriting the delivered record.

`code/09-entire-hahn-holonomic-verification.py` — 2,043 checks, 0 failures. It emphasises the boundary behaviour of the exclusion bounds and the exact theta domain.

Finite calculation	Checks
Theta homogeneous equation, positive and negative valuation	92
Theta first-order coefficient recurrence	45
Dilation and differential forward recurrence conversions	72
Unit orbit: parity, exceptional cancellation, residue polynomial	504
Rank-two noncofinal affine minimum	200
Theta excluded upper-scale argument	200
Theta included coarse boundary	170
Theta cofinal-scale growth	120
Finite-coordinate entire witness growth	480
Sparse recurrence exact identity	50
Escape boundary: constant leading exponent, strict exterior descent	50
Formal exponential recurrence	60
Total	2,043

Two of these rows exist only in this suite and check the two boundary phenomena that the report insists are different: “theta included coarse boundary” checks that a zero coarse valuation coexists with positive growth in the original lower-scale coordinate, which is Section 8.5, and “escape boundary” checks the constant-leading-exponent case that Lemma 2.2 rules out, which is what makes the excluded region in Corollary 3.5 closed. Its `--output` argument is optional. Seeds are recorded in the shipped records.

code/10-nonlinear-rigidity-verify.py — 2,113 checks, 0 failures. It checks the finite algebra of Sections 14–17.

Finite calculation	Checks
Finite Hahn initial layers and exclusion inequalities	450
Automatic first-order nondegeneracy of the corner	240
Univariate falling-factorial corner identities, orders 0 to 3	480
Lexicographic higher-rank ordered-group exclusion bounds	500
Weighted Euler homogeneous-part identities	360
Worked examples: Riccati, degree cutoff, resonance, Painlevé, vanishing corner, Euler solution space	83
Total	2,113

The group names in the record `data/10-nonlinear-rigidity-verification.json` are `finite_hahn_certificate`, `first_order_nondegeneracy`, `higher_order_corner`, `ordered_group_radius`, `weighted_euler` and `worked_examples`, in the order of the table; “radius” there is source 10’s word for an exclusion bound. The default seed is 20260922. The `--output` argument is required, and the program refuses to overwrite an existing file. The vanishing-corner example is tested for the monomial degrees $0, \dots, 20$ only. The counts are program checks, not independently proved theorems.

What the checks do not do. No suite proves any infinite theorem. No finite prefix can prove strong summability of an infinite family, cofinality of (e_n) , noncofinality of $(n\epsilon)$, the existence of an

infinite escape path, the eventual orbit statements, the generic-line theorem, or the absence of every possible annihilating equation. No infinite Hahn sum is evaluated, or numerically approximated, by any of the programs. In particular a finite truncation of the explicit series (11.2) is a polynomial and is D-finite, so no finite computation can witness its nonholonomicity; for the same reason no finite computation can witness the two-jet independence of Corollaries 17.1 and 17.2. Source 10’s suite in particular does not test arbitrary infinite supports, cofinality, all Hahn elements, the nonexistence of all annihilating equations, or historical novelty. What the suites can detect is sign, indexing, conversion and finite valuation-comparison mistakes. Everything else rests on the written arguments.

The mathematical dependency chain is short.

Conclusion	Load-bearing ingredients
Differential rigidity	Integer evaluation lemma; finite shift conversion; escape chain and exclusion criterion.
Unit-dilation rigidity	Residue polynomial; finite binomial leading terms; escape chain and exclusion criterion.
Noncofinal nonunit rigidity	Eventual order of finitely many affine values; noncofinality witness; escape chain and exclusion criterion.
Coarsened exclusion	The same affine values, projected to $\Gamma/\Delta_{ v(q) }$; strictness of the projected descent.
Positive dilation existence	Support monoid certificate and the eventual strict increase of h_n ; or, in the order-unit case, quadratic leading valuations and the cofinal-valuation criterion.
Several-variable rigidity	Countable polynomial avoidance; finite-block regrouping; rational differential system on a generic line.
First-order nonlinear rigidity	Hahn-family algebra and its residue homomorphism; large active degrees; exact corner identity; uniqueness of a corner term for each power of Y' .
Higher-order corner criterion	The same, with $\chi_P \neq 0$ assumed.
Positive-weight Euler rigidity	The same algebra in m variables; finite bounded-weight sets; the eigenvalue identity of ϑ_q .

All supports are sets. All operators are finite sums. All integer indices are ordinary nonnegative integers. The coefficient valuation is trivial in characteristic zero; the coefficient field is $\mathbb{C}$ in Sections 1–13 and any characteristic-zero field in the nonlinear part. The equations use the external formal variable derivative. None of the proofs asserts convergence in the fine topology of $\mathbf{No}[i]$, substitutes a real-valued norm for an arbitrary-rank valuation, or transports a fixed-workspace evaluation to every surcomplex argument.

19 Originality assessment and questions left open

19.1 Established background and proposed contribution

The D-finite/P-recursive correspondence [1], ordinary properties of Hahn fields, the Hahn–Neumann support lemmas and Higman’s finite-word theorem [6], finite polynomial root bounds, the Conway normal-form identification [9, 10], and partial theta series with their elementary functional identity [5] are established material. The companion report’s cofinality and scalar-extension work, including

the group Γ_∞ and the distinction between order units and countable cofinality, is prior project material [12]. The rank-one quadratic-exponent entire example is already in the collection [14]. We claim to originate none of these, and we do not claim to have disproved any published theorem about a different analytic function class.

The proposed research contribution is their combination into the sharp arbitrary-rank dilation classification, Theorem B: in particular the necessity argument for all noncofinal scales, the unit case with root-of-unity residue, the uniform exterior obstruction applying to every solution of a fixed equation, the quantitative exclusion bounds with their two different boundary behaviours, and the exact theta evaluation domain for an arbitrary Hahn parameter. The mixed noncofinal extension (Theorem 10.2), generic-line rigidity (Theorem C), and the explicit series excluded from every nontorsion pure dilation equation (Theorem 11.2) are further consequences proved here.

The nonlinear part (source 10) inherits Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning, and is related to central-index, Newton-polygon, non-Archimedean value-distribution and tropical differential methods (Section 17.4). Its proposed contribution is the precise arbitrary-rank strong-family formulation: Theorem D with its finite exclusion certificate (Theorem 15.1), the candidate degrees and the entire-solution locus, the higher-order corner criterion, the positive-weight Euler theorem, and the non-uniformity of the nonlinear exclusion bound. Its priority is provisional in the sense recorded there.

19.2 A result-by-result attribution ledger

Result or ingredient	Status in this report
Hahn arithmetic, strong support sums, surreal normal forms	Standard background; references and support proofs supplied.
Higman's finite-word theorem	External standard ingredient; alternatively import the classical Hahn–Neumann support lemma directly.
D-finite / polynomial recurrence conversion	Classical; credited to Stanley and derived explicitly for Hahn coefficients.
Partial theta series and functional identity	Classical; not a novelty claim.
Escape chain and exclusion criterion, with the local-finiteness boundary	Elementary support-theoretic lemma and quantitative formulation developed here.
Eventual affine and periodic orbit valuations	Proved here for arbitrary Hahn coefficients; the finite residue-polynomial treatment covers near-torsion units.
Exact fixed- q order-unit classification	Principal proposed contribution: arbitrary rank, arbitrary Hahn parameter, both signs of $v(q)$.
Exact theta evaluation domain	Support-controlled sharpness theorem; related in spirit to the companion report's scalar-extension results, proved independently here.
Uniform exterior obstruction and common degree bound	Proved here; stronger than eventual termination of each solution separately.
Universal single-dilation nonholonomicity on no-order-unit workspaces	Consequence of the classification, with an explicit entire witness.
Mixed differential–dilation rigidity at noncofinal nonzero scales	Proved here; the unit-valuation mixed case is left open.

Result or ingredient	Status in this report
Several-variable rigidity and multivariate algebraic rigidity	Proved here; the generic line is an existence device.
Gauss initial forms, Euler eigenvalues, indicial polynomials, central-index and Newton-polygon ideas	Established (source 10); credited to Hayman, Cano and Fortuny Ayuso, Hu–Luan, Hu–Yang and Giansiracusa–Mereta as predecessors in different settings.
First-order nonlinear rigidity (Theorem D), finite exclusion certificate, candidate degrees and entire-solution locus	Proposed contribution of source 10, over any characteristic-zero coefficient field; priority provisional.
Higher-order corner criterion and positive-weight Euler rigidity	Proposed by source 10; sufficient conditions only; the vanishing-corner case is Question 19.1.
Solution-dependence of the nonlinear exclusion bound	Proved by source 10 with the explicit Riccati family of Theorem 15.10.
Two-jet independence of $\sum t^{n^2} z^n$ and of (11.2)	New consequences (source 10); both series are prior collection material.
Named published open conjecture	None claimed solved; the nonlinear question answered in order one is a question of this report, not a historical conjecture.
Priority, peer review, formal verification	Priority and independent refereeing are not certified. The support word lemma, escape mechanism and formal exponential domain have Lean coverage; the main classifications and the whole nonlinear part remain unformalized.

The targeted literature checks of sources 08 and 09 located the standard D-finite framework, classical complex q -difference growth theory, q -holonomic degree growth in a rational-function setting, and ultrametric q -difference theory, but did not establish whether an identical arbitrary-rank strong-Hahn formulation appears elsewhere. Several broad keyword searches returned irrelevant material and were not treated as evidence of absence; a repository content search for **holonomic** returning no matches is likewise not proof that no sentence anywhere in the repository can be reformulated into one of these results. Accordingly this is a proposed original result with proofs, not a certified claim of bibliographic priority. Independent mathematical review is still needed; machine-checked scope is the partial coverage recorded in Section 13 and the formalization ledger. The corresponding record for the nonlinear part, with its own search limits, is Section 17.4.

19.3 What the nonlinear part does not claim

Every limitation stated by source 10, in its manuscript or its two audit files, is kept; they are collected here.

- (N1) Only the first-order case of Question 19.1 is answered, not the unrestricted higher-order question.
- (N2) Theorem D does not say that a nonpolynomial entire function is differentially transcendental in all orders; independence of f and $D_z f$ does not exclude a relation involving $D_z^2 f$.

- (N3) The criterion $\chi_P \neq 0$ is sufficient, not a complete invariant of a solution set, and the methods do not decide equations of order at least two with vanishing residue corner. Proposition 16.4 is not a nonpolynomial entire solution and does not refute higher-order rigidity.
- (N4) The corner is a finite selection, not a full differential Newton polygon; no invention of differential Newton polygons, no equivalence to tropical differential algebra, and no new tropical fundamental theorem is claimed.
- (N5) The exclusion bound (15.8) is not claimed optimal; the certificate has no converse; leading valuations alone do not decide evaluation at a boundary argument.
- (N6) Candidate degree sets are necessary, not realized; their finiteness is not a uniform algorithm for arbitrary coefficient names, and the workflow of Section 17.3 is not an algorithm for arbitrary surreal names, does not enumerate local formal solutions, does not decide strong-support membership, and accepts no numerical solver failure as a certificate.
- (N7) Corollary 15.6 does not say that the number of solutions is finite, is not a moduli statement over rings with nilpotents, and covers only the extension class it states.
- (N8) No theorem says that G , F or any nonpolynomial series is entire on the proper class $\mathbf{No}[i]$; entireness and the derivative are relative to one set-sized workspace.
- (N9) The independence statements concern formal functions; no statement concerns algebraic independence of a value $f(x)$ from its own ambient field.
- (N10) D_z kills all Hahn coefficients and is not the Berarducci–Mantova derivation; no scalar-derivation result is claimed.
- (N11) The series $\sum t^{n^2} z^n$ and (11.2), and the linear rigidity of the latter, are prior material; only their nonlinear two-jet consequences are new. The tail-span report’s all-order independence results are neither weakened nor rediscovered.
- (N12) Hahn fields, the positive-support summation lemma, Conway normal forms, Gauss initial forms, Euler eigenvalues and indicial-polynomial reasoning are established ideas, not relabelled as new; the positive-support lemma is imported by source 10, and proved in this report by Lemma 2.1 (2).
- (N13) The precise all-rank statements are proposed contributions; historical priority is not certified. The literature search of 22 September 2026 was targeted and not exhaustive; Hayman’s survey and the Hu–Yang monograph were consulted through metadata only, and no claim is made that rank-one special cases were previously unknown.
- (N14) The repository comparison at 048b72c was targeted: no line-by-line audit of every report, Lean file, archive, branch or the Git history, and no absence claim rests on a truncated preview.
- (N15) The answered question is a question of this report, not a named historical conjecture.
- (N16) No Lean formalization and no independent refereeing of the nonlinear part exists. The 2,113 finite checks are program checks, not theorems; they do not test infinite supports, cofinality, all Hahn elements, nonexistence of annihilating equations or novelty, and no infinite surreal sum is numerically approximated. A successful PDF build is not a proof check.
- (N17) The sequence arguments behind the support calculus use ordinary set-theoretic choice.

19.4 Further mathematical questions

Question 19.1 (Nonlinear differential equations of order at least two). Can a nonpolynomial $f \in \mathcal{E}_\Gamma$ satisfy a nonzero algebraic differential equation $P(z, f, D_z f, \dots, D_z^s f) = 0$ over $K_\Gamma(z)$ of order $s \geq 2$? By Theorems D and 16.1, such an f satisfies no nonzero equation of order at most one and no equation with $\chi_P \neq 0$; equivalently, can a nonpolynomial strongly entire function satisfy an equation whose residue corner polynomial vanishes identically, without satisfying any other equation to which Theorem 15.1 applies? Under which hypotheses on Γ can linear rigidity be strengthened to differential algebraic rigidity in all orders? Does the explicit series (11.2) avoid all equations of order at least two under a natural coefficient restriction?

Status. As posed by sources 08 and 09, the question asked about nonlinear algebraic differential equations of every order. Source 10 answers its first-order case negatively, for every nonzero Γ and every characteristic-zero coefficient field (Theorem D): in order one no hypothesis on Γ beyond $\Gamma \neq 0$ is needed, and (11.2) avoids every first-order equation with no coefficient restriction (Corollary 17.2). It also settles the higher-order equations with $\chi_P \neq 0$ (Theorem 16.1) and the relations between f and a positive-weight Euler derivative (Theorem 16.5). What stays open is order at least two with a vanishing residue corner. Proposition 16.4 shows that such equations exist and can have polynomial solutions of unbounded degree, so the study cannot rely on a uniform degree bound for all polynomial solutions of an arbitrary higher-order equation; it is not a nonpolynomial entire solution and gives no negative answer. Source 10 suggests as a next step classifying the polynomial solutions of a vanishing-corner residue equation beyond their leading monomial and determining how those constraints propagate between scales.

The escape proof does not answer this, in any order. A nonlinear differential relation produces convolution sums in the coefficients rather than a bounded-step linear recurrence, and treating it as though it had the form (3.1) would be an invalid extension. The first-order answer uses the different route of Section 14.

Question 19.2 (Beyond a positive Euler operator). Which polynomial vector fields V have the property that every strongly entire f satisfying a nonzero equation $P(x, f, Vf) = 0$ is a polynomial?

Positive integer Euler weights give one family, and zero and mixed weights give explicit counterexamples (Section 16.3). A useful extension would need both a finite-dimensional filtration and a sufficiently nondegenerate action on its highest layers; diagonalization alone is not enough if infinitely many monomials share a bounded eigenvalue. (Source 10.)

Question 19.3 (Effective representations and minimal certificates). For which exact Hahn coefficient representations can the candidate degree set and an optimal excluded valuation cut be computed uniformly?

The bound (15.8) is always a valid choice from exact finite valuation data but can be far from minimal, and Theorem 15.10 shows that an optimal cut must in general depend on solution data. An algorithmic refinement must state how it receives those data and how it decides coefficient cancellation. (Source 10; the linear analogue is Question 19.5.)

Question 19.4 (Mixed operators at unit valuation). For a nontorsion q with $v(q) = 0$, must every strongly entire solution of a nonzero mixed equation (10.1) be a polynomial?

The unresolved point in the present method is uniform control of $v(P(n, q^n))$ for a general bivariate P in the unit case. The pure q -difference proof controls $P(q^n)$ and the differential proof

controls $P(n)$; neither statement by itself handles their combination. This is a question suggested by the argument, not asserted to be a previously catalogued open problem.

Question 19.5 (Optimal exterior regions and exact solution domains). For a fixed differential or rigid dilation operator, can one compute the largest common strong evaluation domain of its formal solution space from finite valued operator data, rather than only a sufficient exclusion bound? Can the coarse exclusion bound of Theorem 7.4 be supplemented by support hypotheses that give the exact domain?

The thresholds supplied here are uniform and explicit once the valuation certificates are available, but need not be optimal, and distinct solutions may have cancellations or terminating tails that change their individual domains. Proposition 5.3 shows that the uncoarsened bound can be exactly attained, and Theorem 8.1 exhibits one sufficient pattern for an exact answer: a common positive support monoid together with a controlled sequence of leading exponents. A full answer should retain arbitrary Hahn tails rather than use only a numerical radius.

Question 19.6 (Several independent dilations). What is the existence criterion for a nonpolynomial strongly entire solution constrained by equations using several multiplicatively independent dilations? Is an appropriate cofinality condition on their joint valuation data sufficient, or do compatibility relations impose additional obstructions? For a no-order-unit workspace, can one classify the finite linear relations among $F(q_1 z), \dots, F(q_s z)$ when the parameters are not powers of a common element?

The single-dilation theorem identifies the exact obstruction when only one scale is repeatedly added. With several dilation directions, distinct monomials can have equal valuation slopes and their leading coefficients can produce nontrivial recurrence phenomena of their own. A meaningful formulation must also quotient out identities caused by repeated parameters and any genuine algebraic relations among them. No multidilation classification is implied by Theorem B.

These questions are directions suggested by the proved statements, not claims that they were already posed in the published literature, and their formulations are deliberately narrower than a general theory of nonlinear surcomplex differential equations.

20 Conclusion

Strong entireness over a surreal Hahn workspace imposes a global scale requirement that a finite linear differential equation cannot meet unless its solution terminates. A dilation equation can meet that requirement, but exactly when its valuation reaches every scale through integer multiples. The decisive distinction is therefore not between Archimedean and non-Archimedean fields, nor between rank one and higher rank; it is between *a cofinal dilation scale* and *a scale trapped below another one*.

The obstruction is finite and robust. A linear recurrence forces a later nonzero coefficient with a controlled valuation, and a large enough argument turns an infinite chain of such coefficients into a nonincreasing sequence of leading exponents. Descending supports and infinite reuse of one exponent are two distinct ways in which the evaluation fails, and keeping them separate is what makes one exclusion boundary closed and the other open. The mechanism works with arbitrary Hahn coefficient tails, arbitrarily high valuation rank, and sparse coefficient sequences, without replacing strong summability by topological convergence and without any hidden completeness assumption.

The countable-rank example makes the separation concrete. The series $\sum_{n \geq 0} \omega^{-\omega^n} z^n$ is evaluable at every point of its full workspace, yet no single nontorsion dilation in that workspace supplies a finite linear equation for it. That gap between existence and finite linear describability is the main mathematical contribution proposed here.

The nonlinear part pushes the same global requirement past linear equations by a different mechanism. A finite algebraic relation between a function and its first derivative is incompatible with nonpolynomial strong entireness over any nontrivial characteristic-zero Hahn workspace: at one large admissible scale the whole series leaves a finite initial polynomial, whose top coefficient is controlled by a corner polynomial that is automatically nonzero in order one. In higher order an exact cancellation of that corner is a genuine obstruction to the method, and a fixed nonlinear equation can then have polynomial solutions of unbounded degree; whether it can have a nonpolynomial entire solution is the open part of Question 19.1.

A Notation

Symbol	Meaning
Γ	A fixed nonzero set-sized ordered abelian group; divisible only where stated.
K_Γ	The complex Hahn field $\mathbb{C}((t^\Gamma))$.
$v(a), \text{lc}(a), \text{res}(a)$	Least support exponent, its coefficient, and the residue when $v(a) = 0$.
$\mathcal{E}_\Gamma, \text{Dom}_{\text{str}}(f)$	Series strongly evaluable at every point of K_Γ ; the exact set where a series is strongly evaluable.
D_z	Formal derivative in z , zero on every element of K_Γ .
σ_q	Argument dilation $f(z) \mapsto f(qz)$.
$M(S)$	The additive monoid generated by a well-ordered $S \subseteq \Gamma_{>0}$.
Δ_β	The smallest convex subgroup containing $\beta > 0$, (2.1).
B, B^+, B_{rec}	Exclusion thresholds in Γ ; B_{rec} divides by the jump length and needs Γ divisible.
$\overline{B}_{\text{rec}}^b, \overline{B}_{\text{rec}}$	Their coarsened analogues in $\Gamma/\Delta_{ v(q) }$.
μ, β	Usually a positive dilation valuation; an order unit only when stated.
$(X)_r$	The ordinary falling factorial polynomial of degree r .
Θ_p	The partial theta series (8.1).
Γ_∞	The increasing-rank direct sum (11.1).
<i>Nonlinear part, Sections 14–17</i>	
$k, \mathbb{K}, \mathcal{E}_\Gamma(k)$	Any characteristic-zero field; $k((t^\Gamma))$; its strongly entire series (Convention 14.1).
$\mathcal{S}_m, \mathcal{S}_m^+, \rho$	Series with strongly summable coefficient family; those with coefficients of nonnegative valuation; coefficientwise residue.
$\gamma_\delta, \phi_\delta, N_\delta$	Gauss value, initial polynomial and top active degree at the scale $t^{-\delta}$, (14.1).
$d_P, w_P, \mathcal{C}_P, \beta_P$	Corner data of a differential polynomial, (14.5).
χ_P, I_P	Residue and full corner polynomials, (14.6).
κ_P	The rational degree cutoff (15.11).
δ_0	A solution-dependent exclusion bound, (15.8).
ϑ_q, wt	Positive-weight Euler operator and weight.

B Proof-dependency and hypothesis audit

The logical dependencies are short. Differential rigidity uses the integer-evaluation lemma, the coefficient conversion, and the escape chain with its exclusion criterion. Dilation necessity uses the geometric-orbit lemma or the unit-orbit theorem, the dilation conversion, and the same criterion. Dilation sufficiency uses only the partial theta identity and the support certificate. The mixed extension uses the bivariate orbit lemma. The several-variable theorem uses countable polynomial avoidance, finite-block regrouping and the one-variable theorem. The infinite-rank application uses a direct strong-entireness proof and the completed classification. The nonlinear part uses none of these linear arguments: first-order rigidity uses the Hahn-family algebra (with the support calculus of Section 2), the large-active-degree lemma, the corner identity and the order-one nondegeneracy of the corner; the higher-order criterion replaces the last step by the hypothesis $\chi_P \neq 0$; the Euler theorem uses the same algebra in several variables with finite bounded-weight sets.

No unproved ring-theoretic conclusion from another report in this collection is used. The real-valued coarsening of [12, Lemma 9.1] is cited and not reproved, and no proof here depends on it. Higman’s theorem is an external standard ingredient; alternatively one may import the classical Hahn–Neumann support lemma directly. The surreal interpretation imports the normal-form field embedding, while every substantive function statement is already proved in the abstract Hahn field.

The potentially dangerous hypotheses, collected for review:

- Supports and the value group are sets; all operators are finite sums; all integer indices are ordinary nonnegative integers.
- $\Gamma \neq 0$, set-sized, ordered abelian. *Divisibility is not assumed* except in Corollary 3.5, the refined threshold (5.4), Theorem 7.4 (b), and Example 13.1, each of which states it.
- Series are ordinary $\mathbb{N}$ -indexed power series in one variable, except in Section 6. Equation coefficients are rational functions of that variable, not arbitrary entire functions; their scalar coefficients may be arbitrary Hahn elements.
- The residue field is characteristic zero and trivially valued. Algebraic closedness is not used. It is $\mathbb{C}$ in Sections 1–13 and an arbitrary characteristic-zero field in the nonlinear part; the latter generality is not transferred to the linear theorems (Convention 14.1). Characteristic zero is essential (Example 17.3).
- Dilation parameters are nonzero and nontorsion; nontorsion is essential, as Section 9 shows by counterexample.
- The derivative is d/dz , not a scalar surreal derivation.
- Entireness is always relative to one fixed workspace, never to $\mathbf{No}[z]$. Strong summability requires both well-ordering and local finiteness.
- The conclusions of Sections 1–13 concern *linear* equations. Nonlinear differential equations are treated only in order one, in higher order under $\chi_P \neq 0$, and for positive-weight Euler relations (the nonlinear part); no nonlinear differential transcendence in all orders, no nonlinear difference transcendence and no unrestricted mixed-dilation independence is claimed.

Two boundary behaviours deserve separate emphasis because they point in opposite directions. The most delicate equality is in Corollary 3.5: weakly decreasing leading exponents already obstruct strong summability, so the excluded region is closed, and Proposition 5.3 shows the bound is attained.

The most delicate strictness is in Theorem 7.4: equal *coarse* values need not obstruct summability in Γ , so that excluded region must be open, and Theorem 8.1 with Section 8.5 shows why.

Deliberate exclusions, recorded so that no reader supplies them by analogy: finite-order q is excluded from the main classification and has genuine periodic-support counterexamples; mixed operators at unit valuation are not classified; no nonlinear differential algebraic rigidity is claimed beyond order one, the nondegenerate higher-order class and positive-weight Euler relations, whose own non-claims are listed in Section 19.3; no complete algorithm for arbitrary surreal comparisons or cofinality decisions is claimed; the finite checks prove no infinite theorem. The main theorem package, the nonlinear part included, has no Lean verification or independent refereeing; the existing support, escape and exponential-domain coverage is recorded in the formalization ledger.

C Reproducing the document and the finite checks

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```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python code/08-finite-recurrences-verify.py --output rerun-08.txt
python code/09-entire-hahn-holonomic-verification.py --output rerun-09.txt
python code/10-nonlinear-rigidity-verify.py --output rerun-10.json
```

The `--output` argument is required by the first and third scripts, so that a rerun need not overwrite the record distributed with the report, and optional for the second; the third refuses an existing output path. All use Python 3.9 or later, exit with a nonzero status if any check fails, and use exact integer and rational arithmetic throughout. The build script `code/10-nonlinear-rigidity-build.py` was written for source 10's own manuscript: run here it would typeset this report, leave auxiliary files in the directory and write `data/build_report.json`. The shipped `data/10-nonlinear-rigidity-build_report.json` and `data/10-nonlinear-rigidity-layout_audit.json` describe the 23-page source manuscript, not this report; use `latexmk` as above, on a copy.

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